DAiR3

DAIR^3

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https://dair-3.org/

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Contents

Introduction

The rigor of scientific research and the reproducibility of research results are foundational to the validity of scientific findings and the trustworthiness of science as an institution. Yet research rigor and reproducibility remain persistent challenges, particularly for work involving complex data types from heterogeneous sources and long data manipulation pipelines. These challenges are especially acute as data science and artificial intelligence (AI) methods emerge rapidly, and researchers work to seize the opportunities that new methods bring while managing the risks they introduce.

The Data and AI Intensive Research with Rigor and Reproducibility (DAIR^3) program is a multi-university initiative funded by the National Institute of General Medical Sciences (Award 5R25GM151182). The program equips faculty and technical staff in the biomedical sciences with the skills needed to improve the rigor and reproducibility of their research, and supports them in transferring those skills to their own trainees.

This curriculum document accompanies the DAIR^3 summer bootcamp series. It is organized into seven thematic units covering the principal competencies of the program.

Curriculum Overview

The seven units of this curriculum address the following themes:

1. **Ethics in Biomedical Data Science.** Foundational ethical frameworks, responsible conduct of research, and the application of ethical reasoning to AI and data-driven decision-making in health contexts.

2. **Data Management, Representation, and Sharing.** Principles of data lifecycle management, metadata standards, data provenance, and practices that support open and reproducible science.

3. **Statistical Foundations.** Core concepts in probability and statistics relevant to biomedical research, with emphasis on rigorous analytical design and appropriate inference.

4. **Predictive Modeling.** Design, training, evaluation, and reporting of AI and machine learning models, including considerations of model transparency and fairness.

5. **Reproducible Workflows.** Version control, computational environment management, and workflow automation as tools for ensuring that analyses are transparent and reproducible.

6. **Meta-Analysis and Assessment Across Studies.** Methods for synthesizing evidence across heterogeneous studies, including systematic reviews and quantitative meta-analysis.

7. **Large Language Models.** Principles of generative AI, prompt engineering, and the responsible integration of large language models into biomedical research workflows.

The 2026 Data Challenge

Each cohort of DAIR^3 participants engages with a data challenge that provides hands-on experience applying the skills developed across the curriculum to a real-world biomedical problem. The 2026 data challenge focuses on **vital statistics**: projecting the prevalence of underweight newborns and infant mortality by county in Texas through the year 2030. This challenge draws on publicly available natality and mortality data and requires participants to integrate skills from data management, statistical modeling, and reproducible workflow development.

Detailed specifications for the 2026 data challenge, including data sources, analytical requirements, and deliverable formats, are provided in the companion Data Challenge document.

How to Use This Document

Each unit in this curriculum follows a consistent structure. An introductory overview describes the scope and motivation of the unit. Learning objectives state the competencies that participants are expected to develop. The instructional content is organized into sections corresponding to individual sessions or topics. Each unit concludes with assessment instruments that participants complete to demonstrate mastery.

This document is produced in two versions: a classroom version intended for use during instruction, and a datathon version distributed to participants as a reference during the data challenge. Certain sections and annotations differ between versions.

Participants are encouraged to engage with the material critically, bring questions from their own research contexts, and share knowledge with their peers. The program is designed as a collaborative learning experience across institutions and disciplines.

Chapter 1

Unit 1: Responsible Conduct of Research

Total Time: 3 hours

1.1 RCR in the Context of Biomedical Data Science

This lesson examines the sociotechnical and ethical aspects of biomedical data science. We will consider ethical issues in the responsible conduct of research that are novel to or pose new challenges in the context of biomedical data science such as reproducibility and privacy. Students will also consider biomedical data science as a sociotechnical system and define roles for themselves and other key constituents.

1.1.1 Learning Objectives

1. Explain novel ethical issues in responsible conduct of research for data science such as reproducibility and privacy.
2. Describe the landscape of biomedical data science as a sociotechnical system and articulate roles.

1.1.2 Assessment Instrument

- Compare responses to the data challenge with your peers. What issues arise?
- Why is the date and time of birth no longer recorded after 1987, only the week of birth? Discuss the privacy implications.
- Identify at least three ethical concerns when projecting underweight newborns and infant mortality by county. Consider: stigmatization of specific counties or populations; secondary use of data originally collected for other purposes; and potential biases in historical data collection.

- Who decides what data to collect, how to store it and how to access it? What biases could there be in the data (e.g., data collection in rural areas, existence of infrastructure)? What is the difference between bias and trend in these data? Provide examples.

1.2 What are Ethics? Ethical Issues in Biomedical Data Science

This lesson equips students to address ethical challenges in biomedical data science. Learners will identify strategies for ethical secondary data use, analyze engagement approaches, and develop frameworks for ethical project review, emphasizing anticipatory governance and responsible data science practices. Case studies will be used that draw on the group project selected for the 2026 cohort.

1.2.1 Learning Objectives

1. Differentiate between traditional bioethical, sociotechnical, and other ethical approaches to data science research and applications.
2. Evaluate key ethical challenges in biomedical data science.
3. Identify and formulate approaches to address ethical issues in secondary use, including anticipatory governance principles.
4. Develop a framework for ethical review of biomedical data science projects.

1.2.2 Assessment Instrument

- Suppose you are doing a study using NCHS vital statistics to develop an AI/ML application that will help policymakers make decisions about allocating resources for prenatal care. Who are the key stakeholders that should be involved in the design, dissemination, and/or evaluation of the application? Develop a stakeholder engagement strategy for presenting your findings to affected communities.
- You have been tasked with developing a draft governance framework for the use of predictive models in guiding decisions made by the State of Texas. Develop a framework to help the State by describing the ethical considerations and key questions and/or recommendations for approaching each ethical issue:
- Accompany your framework with a brief reflection (one paragraph) on how well it addresses key issues. Are there any issues that are missing (e.g., for individuals, communities, or populations)? How well does it anticipate future ethical challenges? What feasibility issues and resource constraints should be taken into consideration if the framework is implemented?

Topic	**Ethical Issue(s)**	**Approach**
Stakeholder identification and engagement processes		
Decision-making structures and authorities		
Monitoring and evaluation mechanisms		
Benefit-sharing policies		
Procedures for addressing unexpected impacts		
Informed consent and/or notification		
Data quality and bias		
Privacy		
Other (please specify)		

1.3 Evaluation Rubric

Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
1. Peer Comparison & Issue Identification (10%)	Thoughtfully compares responses with peers and identifies multiple substantive issues; demonstrates critical reflection on discrepancies and their ethical or scientific implications; connects issues to broader reproducibility and data quality concerns.	Compares responses and identifies several relevant issues; discussion is mostly substantive with some reflection on implications; connection to reproducibility or data quality is partially developed.	Identifies some issues from peer comparison but analysis is surface level; limited reflection on why discrepancies matter for data science practice.	Minimal or no meaningful peer comparison; issues identified are vague or incorrect; little evidence of critical engagement with the data challenge.
2. Privacy & Historical Data Context (10%)	Provides a well-reasoned explanation for why date and time of birth is no longer recorded after 1987; clearly articulates the privacy implications of this change; situates the decision within broader sociotechnical or policy contexts and connects it to data availability and reproducibility.	Offers a reasonable explanation with some discussion of privacy implications; demonstrates solid understanding of how institutional or policy factors affect data collection, with minor gaps.	Provides a basic explanation and acknowledges privacy implications but lacks depth; recognizes that external factors influence data collection without fully analyzing their consequences.	Explanation missing or purely speculative; privacy implications not addressed; no grounding in sociotechnical or historical context.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
3. Ethical Concerns in Projection & Secondary Data Use (15%)	Identifies at least three well-developed ethical concerns specific to projecting underweight newborns and infant mortality by county; thoughtfully addresses stigmatization, secondary data use, and historical bias; demonstrates nuanced understanding of how each concern could affect communities and research integrity.	Identifies at least three ethical concerns with reasonable development; addresses most of stigmatization, secondary use, and bias; analysis is mostly substantive with minor gaps.	Identifies two or three ethical concerns but treatment is uneven or generic; at least one of stigmatization, secondary use, or bias is addressed, though not fully analyzed.	Fewer than two concerns identified, or concerns are vague and underdeveloped; key issues such as stigmatization, secondary use, or bias are absent or incorrectly framed.
4. Data Governance, Bias, & Trends (10%)	Clearly articulates who decides what data to collect, store, and access; identifies multiple plausible sources of bias with specific examples; provides a nuanced, accurate distinction between bias and trend supported by concrete examples from the data.	Identifies key decision-makers and discusses bias sources with reasonable examples; distinction between bias and trend is mostly clear and accurate, with minor gaps.	Names some stakeholders but analysis is incomplete; identifies at least one source of bias with a limited example; distinction between bias and trend is attempted but imprecise.	Fails to address who governs data collection or access; bias sources missing or incorrectly described; no meaningful distinction between bias and trend; examples absent or irrelevant.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
5. Stakeholder Identification & Engagement Strategy (15%)	Identifies all relevant stakeholders across design, dissemination, and evaluation phases with well-justified roles; proposes a specific, realistic engagement strategy for affected communities; anticipates conflicts of interest and addresses them thoughtfully.	Most stakeholders identified with reasonable engagement approaches across most phases; strategy is mostly realistic with minor gaps in equity considerations or conflict of interest analysis.	Key stakeholders named but engagement strategies are generic or limited to one phase; strategy present but lacks specificity or justification for community engagement.	Stakeholders missing or poorly identified; no coherent engagement strategy; shows limited understanding of who should be involved or why.
6. Governance Framework: Ethical Issues & Approaches (20%)	Framework addresses all required areas (decision-making, monitoring, benefit-sharing, unexpected impacts, consent, data quality/bias, privacy); ethical issues and approaches are specific, well-reasoned, and internally consistent; integrates anticipatory governance principles throughout.	Most required areas addressed with reasonable depth; ethical issues and approaches are mostly coherent; anticipatory governance referenced and largely applied, with minor gaps or inconsistencies.	Several required areas addressed but treatment is uneven; some ethical issues identified without substantive approaches proposed; anticipatory governance mentioned but superficially applied.	Multiple required areas missing or poorly addressed; ethical issues vague or incorrect; no meaningful application of anticipatory governance; framework lacks coherence.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
7. Reflective Analysis & Feasibility (15%)	Reflection is incisive and well-structured; clearly identifies gaps in the framework at individual, community, and population levels; thoughtfully evaluates how well it anticipates future ethical challenges; provides specific, realistic assessment of feasibility constraints and resource considerations.	Reflection addresses most required elements with reasonable depth; gaps identified at most levels; future challenges and feasibility mostly considered, with minor omissions.	Reflection is present but uneven; gaps or future challenges acknowledged without substantive analysis; feasibility addressed only superficially.	Reflection missing or minimal; no meaningful identification of gaps or future challenges; feasibility not considered; little evidence of critical self-assessment.
8. Overall Framework Quality & Integration (5%)	Framework is coherent, well-structured, and realistic; all components integrate into a unified governance plan; writing is clear and precise; demonstrates thorough command of course concepts across both sections.	Framework is mostly coherent and feasible; components are largely integrated; writing is clear with minor lapses; good command of course concepts throughout.	Framework is partially coherent; some components feel disconnected or underdeveloped; writing is adequate; course concepts applied unevenly.	Framework is incoherent, infeasible, or largely incomplete; components do not integrate; writing is unclear; limited evidence of course concept mastery.

Chapter 2

Unit 2: Data Management

Total Time: 3.5 hours

2.1 Data Collection and Storage

Time: 0.75 hours

This lesson enables learners to design robust, ethical research plans and identify the optimal data collection method for high-quality, reproducible findings.

2.1.1 Learning Objectives

1. Distinguish between qualitative and quantitative data.
2. Apply methods for gathering, organizing, and cleaning data to ensure quality and integrity.
3. Ensure ethical compliance of data collection, handling, and storage.

2.1.2 Assessment Instrument

- From the Data Challenge, are the more than 100 data elements collected by the National Center for Health Statistics qualitative or quantitative?
- Were the SEER data organized and cleaned in a manner that led to your ability to reliably use them for secondary analyses? Why or why not?
- Why do you think they no longer report date and time of birth, and now only require week of birth?

2.2 Metadata - Data About Data

Time: 0.375 hours

This lesson explores the essentials of metadata in biomedical research datasets, covering data collection methods, population, and context. Students will learn to identify high-quality metadata that supports reproducibility and distinguish it from inadequate metadata, ensuring robust and reliable research outcomes.

2.2.1 Learning Objectives

1. Understand standard components of metadata on biomedical science research datasets: how the data were collected, on what population, under what circumstances, etc.

2. Learn to distinguish between good and bad metadata for reproducibility.

2.2.2 Assessment Instrument

- List 5 critical metadata categories that researchers need to know when reproducing findings.

- From the Mathew E. Hauer article called, "Data Descriptors: Population projections for U.S. Counties by age, sex, and race," what metadata are provided about how the population data were collected?

2.3 Data Representation

Time: 0.375 hours

This lesson examines how data can be represented in multiple ways, highlighting that each representation impacts task efficiency. Students will learn to select optimal data representations tailored to specific research tasks, balancing ease and complexity.

2.3.1 Learning Objectives

1. Understand that the same data can be represented in many ways.

2. Appreciate that each representation choice makes some tasks easier, but others more difficult.

3. Learn how to choose a good representation for the task at hand.

2.3.2 Assessment Instrument

The NCHS data are provided as a flat file with more than 100 variables. What is an alternative representation of these same data? Is the original flat file or your alternative schema more conducive to analyses, and why?

2.4 Data Sharing

Time: 2 hours

2.4.0.1 Data Sharing 101

This lesson introduces the principles of Open Science, focusing on the NIH Data Management & Sharing Policy's rationale and key components. Students will explore the FAIR Guiding Principles (Findable, Accessible, Interoperable, Reusable), learning their definitions and practical examples to promote transparent and reproducible research.

Learning Objectives:

- Appreciate the foundations of Open Science
- NIH Data Management & Sharing Policy: Rationale and Key components
- FAIR Guiding Principles: Definition and examples

Assessment: Define rationale behind NIH Data Management & Sharing requirement; list 5 key components you should include in your 2-page Data Management Plan; list the 4 FAIR principles.

2.4.0.2 Data Sharing - The Reality

This lesson examines privacy and confidentiality concerns in Open Science and data sharing. Students will learn to differentiate between biomedical research types: bench science, human clinical trials, and animal models, and understand the unique data sharing implications for each, including ethical considerations and strategies to protect sensitive data while promoting transparency.

Learning Objectives:

1. Learn about privacy/confidentiality concerns related to Open Science and data sharing.
2. Articulate difference in types of biomedical research (bench science, human clinical trials, animal models) and what implications data sharing has for each.

Assessment:

- Create a 2-page Data Management and Sharing Plan following the NIH requirements for your analyses of the birthweight data challenge.
- Apply the FAIR principles (Findable, Accessible, Interoperable, Reusable) to your datasets.

2.5 Evaluation Rubric

Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
1. Qualitative vs. Quantitative Data (10%)	Accurately classifies all or nearly all of the 100+ NCHS data elements as qualitative or quantitative; provides clear, specific examples from the dataset to justify classifications; demonstrates nuanced understanding of the distinction and its implications for analysis and reproducibility.	Correctly classifies most data elements with reasonable examples; distinction between qualitative and quantitative is mostly clear and accurate, with minor gaps in justification or depth.	Provides a basic classification but with limited examples or incomplete reasoning; distinction is recognized but not consistently or accurately applied to the NCHS dataset.	Classification is missing, incorrect, or conflates qualitative and quantitative data; examples absent or irrelevant; shows little understanding of the distinction.
2. Data Organization, Cleaning, & Quality (10%)	Provides a thorough, well-reasoned evaluation of whether the SEER data were organized and cleaned for reliable secondary use; identifies specific strengths and weaknesses in data organization; connects data quality decisions to reproducibility and research integrity.	Offers a reasonable evaluation of SEER data organization and cleaning; identifies most relevant quality issues with adequate justification; connection to reproducibility is mostly developed, with minor gaps.	Provides a basic evaluation but analysis is superficial or one-sided; acknowledges data quality issues without substantive discussion of their implications for secondary analysis.	Evaluation missing or incorrect; fails to engage meaningfully with SEER data organization or cleaning; no connection drawn to data quality, reliability, or reproducibility.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
3. Ethical Compliance in Data Collection & Storage (10%)	Provides a well-reasoned, nuanced explanation for why date and time of birth are no longer reported; situates the decision within ethical, privacy, and policy contexts; demonstrates sophisticated understanding of how ethical compliance shapes data collection and storage decisions.	Offers a reasonable explanation with some ethical and privacy grounding; demonstrates solid understanding of how compliance considerations affect data collection practices, with minor gaps.	Provides a basic explanation and acknowledges ethical or privacy motivations but lacks depth; recognizes compliance as a factor without fully analyzing its implications.	Explanation missing or purely speculative; no grounding in ethical, privacy, or policy context; shows little understanding of how compliance shapes data collection and storage decisions.
4. Metadata – Data About Data (15%)	Identifies 5 critical metadata categories with clear, relevant examples; provides a thorough analysis of metadata from Hauer's article, including how, on whom, and under what circumstances data were collected; differentiates high-quality vs. poor metadata for reproducibility.	Lists most metadata categories with relevant examples; analysis from the Hauer article is solid but lacks some detail; distinction between good/bad metadata is mostly clear.	Lists some metadata categories but with limited examples or partial relevance; Hauer's article analysis is basic; distinction is made at the surface level.	Fewer than 5 categories or inaccurate examples; superficial or missing analysis of Hauer article; shows little grasp of metadata quality for reproducibility.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
5. Data Representation (15%)	Clearly describes alternative data representations and justifies which schema best supports analysis tasks; insightfully compares the flat file to the alternative, with thoughtful reasoning and examples; demonstrates a strong understanding of how representation affects research.	Describes an alternative schema and makes a reasonable comparison to the flat file, with some justification; demonstrates good understanding with minor gaps.	Identifies a basic alternative, but justification/comparison to the flat file is weak or incomplete; limited examples.	Incomplete, irrelevant, or unclear alternatives; little to no justification or analysis of schema versus flat file; shows limited understanding.
6. Data Sharing 101: Open Science, NIH Policy, FAIR (25%)	Thoroughly defines rationale for NIH Data Management & Sharing; lists 5 key components with well-explained relevance; accurately lists and explains all 4 FAIR principles with practical examples for each.	Defines rationale and lists the key NIH components with basic relevance; lists and explains all FAIR principles, with most examples being suitable.	Provides basic rationale and fewer than 5 NIH components; lists most FAIR principles with limited or superficial examples.	Incomplete rationale and components, missing several critical points; fails to list all FAIR principles or provides incorrect explanations/examples.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
7. Data Sharing (The Reality): Privacy, Confidentiality, Types of Biomedical Research (15%)	Creates a complete Data Management & Sharing Plan tailored to the Texas birthweight challenge, following NIH requirements; thoroughly applies FAIR principles to the dataset; clearly articulates privacy concerns and differences among bench, clinical, and animal model research, with thoughtful analysis of sharing implications.	Plan meets most NIH requirements; applies FAIR principles with minor gaps; reasonably discusses privacy and research type differences.	Plan basic or incomplete; applies FAIR principles at the surface level; discusses privacy/research types with limited detail.	Missing/incomplete plan; fails to address FAIR principles, research types, or privacy concerns meaningfully.

Chapter 3

Unit 3: Rigorous Statistical Design

Total Time: 5 hours

This is a practical introduction to conducting rigorous data-driven research in a health-science setting. Our goal is that upon mastering this section, you will be able to use analytically sophisticated methods to obtain scientifically meaningful, reproducible, and innovative results in your research endeavors. We will focus here on methods for data analysis that can be utilized in the setting of population health research, leveraging official statistics or other systematically collected data with spatial and temporal structure.

This document focuses on design, strategy, and interpretation, and includes no code or numerical results. This Python notebook provides some model analyses from which you can borrow ideas as you start to implement analyses informed by the principles covered here.

3.1 Principles of Study Design for Empirical Research

In this subunit, we will consider how you can develop an understanding of the main ideas in a scientific domain in which you intend to conduct research, how to understand and communicate about the capacity of a given dataset for addressing research topics of interest, and how to develop a specific and tractable research aim.

3.1.1 Learning Objectives

1. You will be able to rapidly internalize the current state of scientific knowledge about a specific health-related topic. The goal here is not to master every detail that a domain specialist would know, but rather to develop fluency with the key known mechanisms, to recognize the quantitative strengths of established relationships, and to identify gaps in the current state of knowledge.

2. You will be able to rapidly assess the structure and capacity of a dataset for conducting research on a specific health-related topic. This includes identifying the units of analysis (who is being measured) and the variables or attributes (what is being measured), understanding how the units of analysis were selected, what population they represent, how the variables were measured, and what types of measurement errors may be present.

3. You will be able to propose creative research aims that contribute to scientific knowledge, and that are tractable to address with available data and methods.

4. You will be able to use appropriate terminology from epidemiology and biostatistics to discuss possible research studies relating to a given health-science domain, that can feasibly be conducted using a provided dataset.

5. You will be able to identify plausible exposures, outcomes, and control variables in a research design.

6. You will be able to communicate in both speech and writing about a data-driven scientific inquiry in a health science setting. This will include strategies for effectively communicating to different audiences using precise but accessible language.

3.1.2 Introduction to birth weight as a biological variable

Birth weight is almost universally measured for all births in developed countries, although globally birth weight is only captured for around 60% of births. In the United States, birth weight is required to be entered on birth certificates in all states, and is therefore essentially always recorded. Birth weight is an important indicator of fetal nutrition, and is a predictor of many short term, medium term, and long term health outcomes. Nevertheless there has been some debate about the meaning and importance of birth weight. Moreover, the interpretation of birth weight is complicated by the fact that it varies by sex, race, gestational age, and birth plurality, and has changed systematically over time.

Low birth weight is associated with several immediate risks for the newborn including risk of infection and mortality, as well as longer-term risks for developmental delays and chronic health conditions in adulthood. Low birth weight is formally defined as the birth weight being less than 2500 grams, and very low birth weight is formally defined as the birth weight being less than 1500 grams. Like most thresholds, these are somewhat arbitrary, and there is strong evidence that quantitative differences in birth weight have health implications beyond the birth weight categories determined by these thresholds.

There are many known causes of low birth weight including pre-term birth (low gestational age), maternal health factors such as undernutrition and infection, low or high maternal age, genetic factors, and issues affecting the placenta. These are possible "direct causes" of low birth weight, but in addition one could look at factors that make these direct causes more likely, such as poverty and access to medical care. In addition, one could look to characterize geographic, temporal, or other possible predictors of low birth weight that may be several steps removed from any direct causal role.

3.1.3 Introduction to the birth weight dataset

The data used here are a complete record (a census) of all live births in the United States in specific years. This is *administrative data*, collected by state health departments, that was subsequently repurposed for research. The sample sizes are large, for example in 1971 there were around 1.8 million live births.

The US National Center for Health Statistics (NCHS) released individual birth records with detailed parental information and geographic location (state and county) until 1988. Due to increasing concerns about data privacy, NCHS has gradually restricted access to detailed data. For example, between 1988 and 2005 geographic information was available only for large metropolitan areas, and after 2005 the geographic information was removed entirely. To facilitate analyses that consider individual and community factors, we will work with the older datasets here.

The raw data and documentation are available from the NCHS here, or directly from this page. You can use the prep.py script to download the data and generate the CSV files that are used in the analyses below.

Each record (observation) in the dataset corresponds to one birth. Many of the variables are self-explanatory but here is some more information about several of the variables:

- Birth weight is the weight in grams of the newborn infant
- Plurality is the number of infants born in a single pregnancy, e.g. 1 for single births, 2 for twins, etc.
- Birth order is the total number of births to a mother, including the current birth, e.g. 1 for the first born, 2 for the second born, etc.
- Birth interval is the length of time (in months) to the previous birth, and is missing for first born infants.
- State and county indicate the geographic location where the birth occurred. These are integer codes that are not necessarily consistently coded between years.

3.1.4 Rigorously framing a study involving the birth weight dataset

Clinical researchers, including specialists from epidemiology and biostatistics, have developed terminology and principles to facilitate rapid and consistent understanding of study designs, encompassing issues relating to aims, data, and methods.

Here, we review some terms and concepts that apply specifically to the birth weight data.

The *unit of analysis* is one newborn baby. Each observation in the dataset pertains to one such baby (multiple births, such as twins, produce multiple observations, one for each newborn). Each baby is characterized in terms of measured variables as discussed above. Several of these variables are *quantitative*, and as such are measured in physical units (grams for birth weight, months for birth interval). Several other variables are *ordinal* (plurality and birth order) – they are represented as integer counts. State and county are *nominal* (or *categorical*). These variables are encoded as numbers, but the numbers convey no quantitative meaning, they are just labels for an unordered, finite collection of spatial entities, and as noted above are not always consistently coded between years.

The data are a *census*, in that essentially all births are included – thus our data constitute the entire population of interest. More commonly, data encountered in research are a subset of the population of interest (the subset can be various types of random sample, or a convenience sample). Much of the field of statistics deals with the issue of inference from the *sample* to the *population*, which seems difficult to apply when the data are a census. We can work around this difficulty as follows, and still have the opportunity to employ conventional statistical thinking in our approach. We can take the perspective that the data we observe for a given year, although it is a complete census of all births, was a random sample from a *superpopulation*, and if we could (counterfactually) revisit that year from an independent vantage point, we would theoretically be able to obtain an independent census reflecting the same point in history. From this point of view, the census becomes a sample from the superpopulation, and the conventional aim of statistical inference as generalizing from the sample to the population is maintained.

The birth weight data only support research that is *observational* in nature. This is because there was no *intervention* (i.e. a treatment) in the study that was systematically administered to some subjects and withheld from others (who would be *controls*). We can consider exposures such as maternal age that may predict or even cause changes to birth weight, but like all predictive factors in these data, maternal age is endogenous – there is no randomization or researcher control over maternal age. This makes it much more difficult to ascribe causality to any findings we make. Nevertheless, there is a rich array of meaningful and rigorous analyses that can be performed, as we will discuss below.

Some of the variables that may predict birth weight are *modifiable* while others are not. For example, the government could pursue policies to reduce the number of births to very young mothers, effectively increasing maternal age. The rate of teenage motherhood has declined dramatically in recent years, demonstrating that this factor is modifiable, although we do not know precisely which factors or policies are the root causes of this decline. As another example, if we consider geographic differences (e.g. between counties), we may see that counties with better health care infrastructure or higher educational attainment have a lower rates of preterm or low birth weight births. Differences associated with geography are at least in principle subject to modification, since it is circumstances associated with the geographic regions, rather than the regions themselves (i.e. their latitude and longitude coordinates) that cause the differences. As one more example, birth order is related to family size, which changes over time due to social and cultural changes. It is modifiable in theory, by restricting family sizes, but it is hard to see this being acted upon in the United States.

The data we have here are *spatial* and *temporal*. However they are not *longitudinal*. Each birth took place in a given known year, in a specific US county, both of which are recorded. If we combine data from several years, we can consider changes over time in aggregated summaries, for example, increasing mean birth weight. The data are not longitudinal because the units (the babies) are not measured repeatedly over time.

The data are *partially observed*, that is, there is some missing data. The most commonly missing data is information about the father. These missing values are unlikely to be missing "at random" because births in which the father is not available are likely to be systematically different from births in which the father is present. Many factors, both measured and unmeasured, may covary with the availability of information about the father. This *informative missingness* can give rise to methodological challenges which we will return to later.

The data are *administrative*, as they were collected due to the interest of the government in documenting the population, but not for any specific research purpose. They were subsequently repurposed for research. The data were collected *prospectively*, in that information about each birth was collected at the time of the birth, and did not depend on anyone recalling circumstances of the birth from a later point in time (in which case the data would be *retrospectively* collected). However research involving these data would be considered retrospective, as hypotheses and research aims would emerge after the data were available for exploratory analyses to be conducted.

The data are *clustered*, or *multilevel*, which means here that there is a grouping of the births into subsets, determined by the county in which each baby was born, such that babies born in the same county will tend to have similar attributes. The strength of clustering may be weak for most factors, but there is still some such tendency. Clustering likely reflects unavailable *stable covariates* that are constant for all births in a county, such as unmeasured social or environmental factors. For rigorous analysis, clustered data should be analyzed using methods that account for the clustering, otherwise uncertainty is likely to be understated.

Fortunately, the data at hand are unlikely to be subject to large *measurement errors*. Birth weights are recorded on birth certificates and constitute important vital records, so the clinicians collecting birth data are likely to be quite careful. However in a dataset of over a million observations, obtained in an era when the data were likely initially recorded manually on paper, there are likely to be some transcriptional errors. There could also be deliberate falsification of some data (e.g. a parent misreporting their age). We have no way to quantify how frequently this occurs, but presumably it is quite rare. By 1971, most births took place in hospitals, and birth weight should usually be captured with a standard medical-grade scale, but since the data are obtained in thousands of facilities across the United States, quantitative measurement errors may be present, e.g. due to scales being miscalibrated or faulty, but again it is likely that most birth weight measurements are quite accurate.

Parental races are available, although we will not use that data in this unit. Race is to a large extent a construct, and in 1971 attitudes about race were very different from what they are today. Moreover, even with the best intentions, it is impossible to fully capture a person's ancestry in a categorical variable. Furthermore, differences associated with a race construct are unlikely to be due to the ancestry itself, but rather to social and cultural factors that co-occur with race.

3.1.5 Research aims

One of our learning goals is that you will be able to develop a research aim grounded in the current state of knowledge of a health science topic, which here will be the epidemiology of birth weight. We will provide you with quite a bit of scaffolding before asking you to come up with your own research aim in the assessment below.

Data scientists typically work in collaboration with health science "domain experts", with the latter often being primarily responsible for proposing hypotheses and aims, while the former are responsible for identifying analytic methods, and for carrying out the analysis. Interpreting results and reaching conclusions is usually a joint effort. However, these "role boundaries" are blurry and subject to change as tools become available that lower the bar for both the conceptual and technical aspects of research. Here, we will put you in the position of developing a research aim on your own, grounded in the brief overview of birth weight provided above.

We will not tell you what your research aim should be, instead we will provide you with the scaffolding needed for you to develop your own science-driven research aim that can be addressed using the NCHS birth weight dataset. Your research aim does not need to center on a single binary hypothesis, you are welcome to propose a somewhat more open-ended and exploratory aim. But there should be a limited and explicitly defined scope for your research aim, and it should be grounded in a clear conceptual framework.

To provide you with some ideas that may prompt your creative thinking, you may wish to consider the high level goal of understanding the associations between risk factors and birth weights. These risk factors could include maternal and paternal ages, offspring sex, plurality, birth order, parental races, and birth interval. Other productive directions would be to consider systematic geographic and/or temporal variation in birth weights. In your first assessment, you will be asked to develop a research aim that will form the basis of your subsequent work in this unit.

When developing a research aim around risk factors for an outcome, it is very natural to think in terms of approaches involving *regression analysis.* However it is important to note that regression analysis is the method used to achieve the aim, rather than the aim itself, i.e. your scientific aim should not be to conduct a regression analysis. We will discuss regression analysis in more detail in section 3.2, but briefly, what we mean here by "regression analysis" is broadly any method that seeks to learn from data something about the conditional distribution of an outcome Y given predictors X. This includes not only least squares and linear regression (where the focus is on learning the conditional expectation of Y given

X), but also any other quantitative and algorithmic approach that estimates some facet of the conditional distribution of Y given X, e.g. a conditional variance or a conditional quantile. This includes a wide array of regression methods, most of machine learning, and a substantial part of what is now known as AI.

While some meaningful research aims can center on a goal of making predictions, thinking mechanistically about scientific aims inevitably involves thinking about causes. Arguably, if you are not thinking about causes, you are not doing science. Therefore, we discourage you here from having a research aim that, for example, aims to achieve high predictive accuracy for the outcome (e.g. birth weight). Such prediction-focused research aims can be useful in practice, but generally provide little insight about the fundamental workings of the system at hand.

3.1.6 Statistical models and causal diagrams in mechanistic research

We assume that you have a basic understanding of regression analysis, which broadly speaking encompasses any method for quantitatively understanding the relationship between explanatory variables and an outcome. Conventional regression analysis views the data in a very "flat" manner. There are only inputs (the predictors or explanatory variables) and an output (the outcome or response). Modern epidemiology favors thinking in terms of causal diagrams, which allow for multiple levels of structure. In section **??** we will propose a possible causal diagram for birth weight and some of its predictors. But before proceeding to that, we will first provide a brief overview of some ways to deepen your understanding of quantitative relationships by thinking beyond an "input/output" view of regression analysis.

- In a *mediation analysis*, an exposure X predicts a mediator M, which in turn predicts an outcome Y (providing an *indirect effect* of X on Y through M). At the same time, X can have *direct effects* on Y that do not involve M.

- In a *moderation analysis*, the relationship between an exposure X and outcome Y is not the same for all units (i.e. for all patients or subjects), but rather is *modified* by a third variable Z. This allows us to think in more personalized terms about both exposures and interventions. When X represents a treatment, this framework permits identification of *heterogeneous treatment effects*. In data observed over space and time, we can aim to identify temporal and/or spatial heterogeneity, sometimes referred to as *nonstationarity*.

- In longitudinal data, we can develop models that consider differing roles for a predictor based on when it was measured relative to the outcome. For example, there can be delayed or cumulative effects of a risk factor, as well as effects that reflect the risk factor as measured concurrently with the outcome.

- Sometimes the explanatory variables can be partitioned into those denoted X that potentially cause the outcome Y, and those denoted Z that are caused by the outcome Y and by the causal predictors X. In this case, Z is known as a *collider*, and it is usually misleading to consider Z as a predictor of Y, or to control or adjust for Z in the analysis.

- The explanatory variables may be partitioned into those that are modifiable, denoted X, which are of special interest as the targets of potential intervention. Other explanatory variables, denoted Z, are immutable, or like the passage of time are not subject to intervention. The latter variables may be mainly of interest as *precision variables* that explain some variation in Y and make it easier to identify the roles of the potentially modifiable factors X.

- The explanatory variables may be partitioned into those denoted X that are modifiable or otherwise of primary interest as potential causes of the outcome Y, while other variables Z are *common causes* or *confounders* of the relationship between X and Y. These confounders may be observed, in which case various adjustments may be made to attenuate their effects, or they may be unobserved, but nevertheless remain conceptually important when developing the research aim.

Causal and mechanistic thinking opens up many opportunities for achieving insight, but is also fraught with challenges, especially when working with observational data, as we have here. By *observational data*, we mean that the potential causes of birth weight (e.g. maternal age) are not assigned as part of the research study (e.g. by randomization), but are instead "selected" by the subjects themselves. Such factors can also be referred to as *exogeneous*. This introduces major challenges into the process of conducting rigorous research, and gives rise to the very active methodological domain of *causal inference*. We will not provide a comprehensive overview of causal inference here, and will limit ourselves to just a few key observations.

First, we think it is counterproductive to throw up our hands and state that due to the challenges of discussing causality, we will limit ourselves to discussing predictive relationships and characterize all findings as *associations*. This generally will lead to our work having limited scientific impact. It is important to engage with mechanistic and causal thinking, while acknowledging the limitations of our conclusions when working with non-experimental data.

For the purposes of this workshop, we can put ourselves in the position of being modelers who have the opportunity to examine rich, high-quality datasets using a wide array of modern modeling methods and powerful computing resources. We should not be afraid to posit various mechanistic relationships and explore their fitness to the data. While it is important to acknowledge the possibility that a finding could have different interpretations if strong confounding were present, it is not always helpful to repeatedly emphasize the qualifications of our findings that arise due to various causal mechanisms, to the point where this obscures any other messages arising from our work.

3.1.7 An initial causal diagram for the birth weight data

A scientifically grounded research aim should reflect at least a provisional notion of the possible causal relationships among the predictors, and between the predictors and the outcome. We will attempt to characterize these relationships here, emphasizing that this is intended to capture only the strong and essentially undisputed relationships. We restrict ourselves to a subset of the predictors available in our extract from the NCHS data (maternal and paternal ages, birth order, plurality, and sex). As an exercise, you could consider how additional factors such as parental races and birth interval could be incorporated into this diagram.

Our provisional causal diagram is shown in Figure **??**. We explain the rationale behind some of our choices next. To begin, all five of our predictors (maternal and paternal ages, birth order, plurality, and sex) have plausible causal effects on birth weight, so there is an arrow in the diagram from each of them to birth weight. It is important to note that these are not all of the possible causes of birth weight, as there are many factors that we do not measure such as maternal nutrition and genetics.

Perhaps the strongest single predictor of birth weight is plurality. For obvious reasons, when multiple fetuses gestate together, each will tend to be smaller than would be the case in a single birth. While plurality is largely random, it is not fully exogenous – several of our other predictors, besides affecting birth weight directly, also affect plurality. For example, it is known that older mothers are somewhat more likely to have multiple births, but no such association is known for older fathers. Also, infants born in plural births are more likely to be female, but it is unclear whether sex causes plurality or plurality causes sex, so we draw this as a bidirectional arrow in our causal diagram. Due to the relationships discussed so far, maternal age is likely a confounder (a common cause) of the relationship between plurality and birth weight.

Offspring sex is determined by whether the fertilizing sperm carries an X or a Y chromosome. This is nearly random, although some research has shown slight deviations from perfect randomness. While it is unlikely that offspring sex causes maternal age, there could be small causal effects of maternal age on offspring sex. Specifically, older mothers are found to have a slightly higher probability of having children of the same sex, but this sex could be either female or male, and there is little evidence of a directional bias in the offspring sex for individual births. This is an example where, even in the context of observational data, mechanistic understanding allows us to rule out strong confounding.

Offspring sex causally influences birth weight, with girl babies being around 100–200 grams lighter than boy babies (as with many mean differences, individual variability is far larger than the mean difference, so many boy babies will be lighter than the average girl baby, and vice versa). Arguably, as offspring sex is nearly independent of other factors, it is an important precision variable, but could only be a very weak confounder of other relationships in our causal diagram.

Maternal age is commonly discussed as a predictor of birth weight, and in particular it has an "inverse U-shaped" relationship in which mothers of intermediate age (around 35 years) have the heaviest babies on average. Paternal age is also

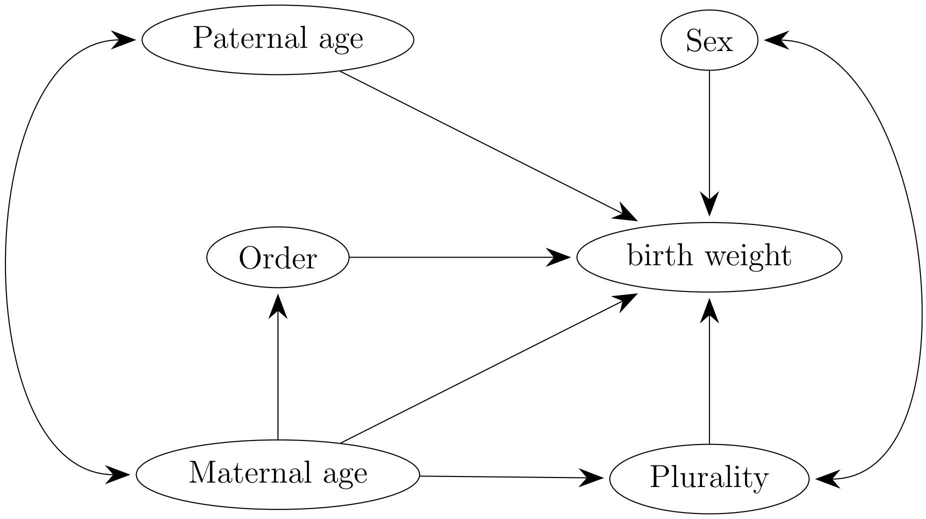

Figure 3.1-1: Causal diagram illustrating possible causal relationships among predictors of birth weight.

a plausible causal predictor of birth weight, although the mechanisms behind this relationship are largely different from those for maternal age – they are potentially less direct, and are likely of smaller magnitude. Due to the relationships between maternal age, plurality, and birth weight, plurality is a possible mediator of the relationship between maternal age and birth weight (it would almost certainly be a partial mediator rather than a full mediator).

Paternal age is positively associated with maternal age, although it is unclear whether this relationship is causal, so we draw the arrow between maternal and paternal age as bidirectional. Complicating matters, paternal age is sometimes missing (maternal age is never missing), and is unlikely to be *missing at random*, introducing yet another source of bias into the analysis if we choose to pursue *complete case analysis*. We could add another variable "paternal age observed" to our diagram, to reflect the possibility that there is information in the missingness status of the paternal age variable. Since paternal age is predictive of maternal age, and is a causal predictor of birth weight, it is likely a confounder of the relationship between maternal age and birth weight (and conversely, maternal age is likely a confounder of the relationship between paternal age and birth weight).

Finally, birth order is causally impacted by maternal age. It is associated with paternal age (through the association of paternal with maternal age), but it seems unlikely that paternal age causes birth order. Birth order likely has its own causal effect on birth weight, but since it is collinear with maternal age, it might be somewhat difficult to distinguish these two effects in the data.

3.1.8 Assessment Instrument

- In one or two sentences, state a research aim that can be addressed using the NCHS birth weight data discussed above. Your aim should be grounded in the current state of knowledge, have a limited and explicitly defined scope, and reflect a clear conceptual framework. The statement of your aim should emphasize what you intend to learn, not the methods you intend to use. Avoid framing the aim purely in terms of prediction accuracy or statistical methods.

- Write a roughly half-page memo to yourself providing a broader consideration of the dataset, study design, and research domain that provides the foundation for your research aim. Your memo should address: the capacity and limitations of the NCHS data for your aim; the relevant causal or mechanistic relationships (referencing the causal diagram where appropriate); the key exposures, outcomes, and potential confounders in your design; and any data quality or missingness concerns that may affect your study.

3.2 Developing an Analytic Plan

In this section, we will discuss how to develop a precise and actionable analytic plan.

3.2.1 Learning Objectives

1. You will be able to develop a rigorous analytic plan to address a stated scientific aim. The analytic plan should be able to be implemented using available data and should employ sophisticated and rigorous analytic methods.

2. You will be able to explain the difference between conditional and marginal distributions, and identify which formal comparisons address a given research aim.

3. You will be able to propose summary statistics that provide initial insight into research questions and that can capture the direction of a relationship, and the magnitude of the relationship in both relative and absolute terms.

4. You will be able to explain the different ways that an auxiliary variable can enter an analysis, such as being confounders and precision variables, and you will recognize when an auxiliary variable is likely to introduce confounding bias.

5. You will be able to engage in a sophisticated discussion of quantitative relationships among measured quantities. This includes considering how such relationships can be assessed and how they contribute to achieving research aims.

3.2.2 Analytic methods

Now that your research aim is stated, we can consider how to approach it analytically. We are using here a dataset that is freely available, which allows us to immediately propose methods for our analytic plan. But it is important to note that in many cases where new data are to be collected or purchased, or if significant obstacles to data access are in place, you would need to provide explicit data collection protocols, and have your analytic plan approved, as well as obtain funding. These are extremely important issues, so to at least partially acknowledge this reality, we will engage here in a discussion of analytic methods without allowing ourselves to immediately jump into exploring the data. That is, we allow ourselves basic information about the data structure, sample size, and measurement units, but do not immediately perform any preliminary analyses, and instead will put ourselves in the position of devising an analytic plan without having access to any data.

Each of you will have your own research aim, but very likely it will involve regression analysis of the birth weights with a focus on identifying and understanding predictors of birth weight, especially low birth weight, employing some level of causal and mechanistic thinking. As noted above, birth weight is often viewed with respect to thresholds, e.g. < 1500 grams is very low and < 2500 grams is low. This may lead one to divide the births into three groups, and you are welcome to do this. However in all the discussion below we will take alternative approaches in which the data are analyzed and interpreted quantitatively.

3.2.3 Descriptive statistics and effect sizes

While regression analysis is likely to be the main tool for your analysis, in some cases you can get a lot of insight by considering summary statistics. This also provides us with the opportunity to begin talking about effect sizes, which is an important component of how scientists quantify relationships in data.

As a simple example, consider the difference in birth weight between girl babies and boy babies. According to the NCHS data, the median girl baby in 1971 weighed 3260 grams and the median boy weighed 3402 grams. The difference between these two weights is -142 grams. This can be viewed as an absolute quantification of the effect size for the sex difference in birth weights. However this number has units (grams) and does not provide a sense of how big this difference is relative to the overall variation in birth weights. To accomplish this we can divide the difference in locations by a measure of scale (or dispersion), which is another descriptive statistic. We can measure dispersion here using the interquartile range, which is 652 grams for females and 680 grams for males. The pooled dispersion is the average of the sex-specific dispersions, which is 666 grams. A standardized effect size is therefore equal to -142 / 666 = -0.21 (which has no units and is thus dimensionless). That is, the sex difference in locations (medians) is around one fifth of the dispersion within either sex.

The median sex difference, while likely important, is much smaller than variation due to other sources. One implication of this is that we will frequently find girls who are born heavier than the typical boy, even if the median girl is born lighter than the median boy. We can quantify this phenomenon with another descriptive statistic, the Kendall's tau concordance. For sex and birth weight the value of this statistic is 0.1, which means that 55% of the time, when randomly selecting one girl and one boy from the population, the birth weight of the boy will be greater than the birth weight of the girl, and 45% of the time the birth weight of the girl will be greater than the birth weight of the boy. This rather modest concordance helps put the effect size of the sex difference in birth weights in context.

Any summary statistic that aims to quantify the magnitude and/or direction of a relationship can be viewed as an effect size. We will not provide a comprehensive review of descriptive statistics and effect sizes here, but wish to introduce this concept now due to the important role that it will play later. Effect sizes play a major role in scientific communication, and also are important in assessing statistical power. The two examples provided above should serve to illustrate descriptive statistics and effect sizes in the context of this research domain.

3.2.4 Regression analysis that focuses on the conditional mean

A natural starting point would be to focus on the conditional mean of birth weight given some risk factors as predictors. The most common approach for achieving this goal is to use generalized linear modeling, which includes linear regression fit with least squares as a special case. As a simple starting point, suppose that we plan to use linear regression fit with ordinary least squares (OLS) to model birth weight in terms of maternal age. Linear regression posits a linear mean structure, and a basic mean structure model would be $E[Y|A_m] = b_0 + b_1 A_m$, where Y is birth weight and A_m is maternal age. However there is strong prior knowledge that low birth weight is more common for very young and older mothers, and less common for mothers of intermediate age. A basic way to accommodate this nonlinearity is to posit the mean structure model $E[Y|A_m] = b_0 + b_1 A_m + b_2 A_m^2$, which is a technique known as *polynomial regression*. Polynomial regression has been used for 100 years or more, but since the 1980's a more general perspective of using basis functions in regression has emerged. While polynomials are one type of basis function, other types such as splines have more favorable properties.

It is important to note that polynomial regression is still considered to be a linear model, since the mean structure is linear in the unknown parameters b_0, b_1, b_2. This simple example illustrates how people often mis-state the way in which linear regression is linear – linear regression does not assume that the conditional mean of the response is a linear function of the predictors, but rather that the mean structure model (or more precisely their estimating equations) are linear in the parameters. This latter type of linearity greatly simplifies estimation and uncertainty assessment of our estimates of the unknown parameters b_0, b_1, b_2.

In the specific case of quadratic (polynomial) regression, the relationship between birth weight and maternal age is convex if $b_2 > 0$ and concave if $b_2 < 0$. The vertex of this relationship is $A_m^\star = -b_1/(2 \cdot b_2)$. While $A_m^\star$ always exists if $b_2 \neq 0$, the value of $A_m^\star$ may lie outside the range of the data, in which case the relationship is monotone throughout the relevant range of ages. Special interest will follow if A_m^* lies inside the realistic range of ages, so that, if $b_2 < 0$, there is a maternal age that maximizes expected birth weight, while expected birth weight declines for either low or high maternal age.

Integrating our discussion so far, it is reasonable to propose the mean structure model $E[Y|A_m, A_p, S] = b_0 + b_1 A_m + b_2 A_m^2 + b_3 A_p + b_4 A_p^2 + b_5 F$, where A_p is paternal age, and F is infant sex coded (arbitrarily) as $F = 1$ for females and $F = 0$ for males. Furthermore, we can safely assume that F is independent of both A_m and A_p, and there is very unlikely to be any moderation (interaction) between F and either A_m or A_p. Moderation between A_m and A_p cannot be ruled out, but we will not consider this possibility here.

While our research questions here relate to the conditional mean structure, we should also consider the conditional variance structure and conditional covariance structure. Variance structure refers to the scatter of the data around the conditional mean, and can be written $\mathrm{Var}(Y|X)$. It is often described using the terms *homoscedastic* and *heteroscedastic*, where homoscedasticity implies that $\mathrm{Var}(Y|X = x)$ is constant in x, and heteroscedasticity indicates that it is not. A complete analysis should consider variance structure in detail, but we will not emphasize that here.

Covariance structure refers to the possibility that birth weights for two different infants can be correlated. Formally, the covariance could be written $\mathrm{Cov}(Y_1, Y_2|X_1, X_2)$, where Y_1, X_1 and Y_2, X_2 refer to two different births. One natural source of covariance structure is sibling relationships, but the NCHS data does not provide that information. Another source of covariance structure would be where due to unobserved covariates, two newborns are likely to share factors in common that we have not measured. For example, two newborns in the same county in the same year are likely to share similar exposures that we have not measured, so their birth weights will be more similar than those of two newborns from different years and/or locations.

3.2.5 Regression analysis that focuses on low birth weight

While mean structure differences are of genuine scientific interest, they do not directly address questions about low birth weight, which is the central interest in many studies of birth weight. There are many regression approaches that target the low (or high) birth weights instead of the conditional mean birth weight, and it is quite interesting to consider the possibility that the causal architecture of the extreme birth weights may differ from that of the conditional mean. Some powerful tools like expectile and expected shortfall regression have been applied in this setting, but here we will focus on the most direct and familiar approach which is quantile regression.

Quantile regression aims to estimate the p^{th} conditional quantile of Y given X, which can be denoted $Q_p(Y|X)$. For example, setting $p = 0.01$ gives us $Q_{0.01}(Y|X)$, which estimates the one percentile point (the weight such that only 1% of newborns are below this weight). Classical quantile regression models the quantiles using linear predictors, analogous to the mean structure models discussed above. For example, we can adopt the model $Q_{0.01}(Y|A_m, A_p, S) = b_0 + b_1 A_m + b_2 A_m^2 + b_3 A_p + b_4 A_p^2 + b_5 F$. It is important to note that the coefficients (b_0, b_1, etc.) depend on the particular probability point being modeled (here 0.01).

Quantile regression is not a GLM, and an important point of contrast between quantile regression and linear regression fit with OLS is that the estimating equations for the unknown parameters are not linear in the parameters, even when the quantile itself is modeled as a linear function of the parameters (as in the example above). This substantially complicates estimation and inference. Although these important technical details are largely handled by available software, it is important to be aware of these complications, especially regarding how inference is conducted, as this will have implications for the power analyses to be discussed later in section **??**. Another challenging aspect of quantile regression that is important to be aware of is that statistical power, and the ability to conduct valid inference, becomes more challenging as the probability point p approaches 0 or 1, which is exactly our regime of interest. This challenge is partially offset by the very large sample sizes that we have available for this study.

We will briefly comment on an analysis approach that while natural, is not further developed here, which would be to categorize every birth into a group such as very low birth weight (< 1500 grams), low birth weight (1500–2500 grams), or normal weight (> 2500 grams). This would lead to using regression approaches such as logistic or ordinal logistic regression (although many other methods are available). On the one hand, this seems to directly address questions around low birth weight, whereas the conditional mean and quantile regression approaches yield results that are more indirect. On the other hand, these category-based approaches are heavily dependent on numerical thresholds which, while widely used, are somewhat arbitrary. Two newborns of weights 1499 and 1501 are treated as completely different, whereas two newborns of weights 1501 and 2499 are treated as the same. A related issue is that much of the information in the data is thrown out by recoding the data into these categories. While you are welcome to use these methods in your work for this unit, we encourage you to carefully consider these drawbacks.

3.2.6 Assessment Instrument

- Develop a brief analytic plan (approximately one half page in length) that addresses the research aim you posed in Assessment 1, using the NCHS birth weight data. The analytic plan should be specific enough that the results would be reproducible if implemented by different researchers.

- Your plan should include: the specific regression or analytic method(s) you will employ and the rationale for selecting them; the outcome variable, key exposures, and any control or precision variables; how you will handle potential confounders, effect modification, or clustering in the data; and at least one descriptive statistic or effect size that provides initial insight into your research question, including both absolute and relative quantifications where appropriate.

- Briefly discuss any limitations of the proposed analytic approach, including assumptions that may not hold and how violations might affect your conclusions.

3.3 Bias, Causal Interpretation, and Statistical Power

3.3.1 Learning Objectives

1. You will be able to explain the concepts of effect size, parameter estimation, and estimation precision, which are the core theoretical ideas upon which *a priori* power analysis is based.

2. You will be able to explain statistical uncertainty, including (a) *a priori* notions of uncertainty as reflected in statistical power, and (b) uncertainty following data analysis as reflected in statistical measures of confidence, significance, and precision. You will be able to identify specific features of the data and methods that result in lower confidence and higher bias.

3. You will be able to explain the conventional definition of power as the probability of research success contingent on the unknown true effect size. You will also be conversant in alternative notions of power that are directly related to estimation precision or prediction accuracy rather than employing hypothesis testing.

4. You will be able to articulate how assessments of statistical power do and do not reflect the real-world reproducibility of analytic findings.

5. You will be able to explain how certain forms of bias can be reduced or eliminated analytically, while others cannot.

3.3.2 Review of the foundations of statistical inference

Statistical inference is concerned with the estimation of parameters that address research aims, and assessing whether the estimated values of these parameters are likely to be close to their true values. This almost always involves invoking the notion that the data are a sample from the population of interest, and that we estimate parameters using the data, but judge their accuracy in relation to the unknown population. Since we do not have access to the population, the estimation error can never be known, but by leveraging our knowledge of the way the data were collected, and considering carefully the properties of the estimation approaches that we use, we can infer meaningful properties of estimators, most notably their bias and their precision.

A parameter estimate is a function of the data, and since the data are random, the estimator is also random. Therefore, the estimator follows a probability distribution that we call the *sampling distribution*. Letting $\hat{\theta}$ denote the estimator, we can write $\theta_0 = E[\hat{\theta}]$ as the expected value of the estimator. The true value of the parameter in the population, also known as the *estimation target* or *estimand*, can be denoted θ. The *bias* is $\theta_0 - \theta$, and the *standard error* of the estimator is $\mathrm{SD}(\hat{\theta})$.

Put in more intuitive terms, bias reflects *systematic error* – the tendency of an estimator to be wrong in the same direction on average over all samples of data from the population. Precision reflects *random error* – the tendency of an estimator to be wrong in different ways for every sample of data that is analyzed, so that these errors average to zero over many independent replications. These two sources of error can be combined to obtain the *total error*, often quantified as the *mean squared error* (MSE), which is $\text{bias}^2 + \text{SE}^2$. For communicating findings, it is more natural to use the *root mean squared error* (RMSE), which is $\sqrt{\text{bias}^2 + \text{SE}^2}$, since this value has the same units as the data. Below we will discuss in more detail what factors may lead to greater bias and/or greater variance in a research study.

Our focus here is on regression analyses, and we can first consider a linear regression model fit with ordinary least squares (OLS). Statistical software will always provide the standard errors for you when you are performing a regression analysis, but it is important to appreciate that the statistical uncertainty in a regression analysis arises from four orthogonal factors, as expressed in the relationship

$$\begin{aligned} \text{SE}(\hat{\theta}_j) &= \sqrt{\text{VIF}_j \cdot \text{Var}(Y|X)/(n \cdot \text{var}(X_j))} \\ &= \text{VIF}_j^{1/2} \cdot \text{SD}(Y|X)/(n^{1/2} \cdot \text{SD}(X_j)). \end{aligned}$$

The terms in this expression can be interpreted as follows:

- $\text{Var}(Y|X)$ reflects unexplained variability or residual scatter in the response variable. The standard error of every regression coefficient is directly related to the standard deviation of this residual scatter.
- n is the sample size. The standard error of every regression coefficient is directly related to $1/\sqrt{n}$ (which is also true of the sample mean, and most other statistical estimators). One implication of this "root n scaling" is that you need to increase your sample size by a factor of 4 to reduce the standard error by half.
- $\text{Var}(X_j)$ is the variance of the j^{th} covariate. The more spread out the values of an explanatory variable are, the easier it is to assess the partial slope for this covariate. The standard error of every regression coefficient is inversely related to the standard deviation of its corresponding covariate. If $\text{SD}(X_j)$ is zero, the standard error of $\hat{\beta}_j$ is infinite and the parameter is not estimable.
- VIF_j refers to the *variance inflation factor*. It is defined to be $1/(1 - R_j^2)$, where R_j^2 is the squared correlation when regressing X_j on the other covariates. The VIF does not involve the response variable Y at all. VIF reflects collinearity, and specifically the fact that it is harder to estimate the coefficient of a covariate if that covariate is itself related to other covariates. For example, if half of the variance of X_1 is explained by regressing it on $X_2, \ldots, X_p$, then the standard error for $\hat{b}_1$ is $\sqrt{2} \approx 1.4$ times greater than it would have been if X_1 were uncorrelated with the other covariates.

There are many important implications of this expression for the standard error of a regression coefficient. For example, we see that the standard error scales with $\mathrm{SD}(X_j)$ and the estimate $\hat{\beta}_j$ scales the same way. That is, if we rescale an explanatory variable by a constant factor (e.g. to change measurement units), both the parameter estimate and its standard error scale by this same factor. Thus, the Z-score, which is the ratio of the parameter estimate to its standard error, is unaffected, and hence is *equivariant* or *dimension-free*. It is also notable that $\mathrm{Var}(Y|X)$ appears in the numerator of the standard error, but $\mathrm{Var}(X_j)$ appears in the denominator of the standard error. Thus, some forms of variation are favorable and others are unfavorable, at least in terms of their impact on estimation precision.

3.3.3 Causal interpretation of findings

In principle, if we specify a statistical model correctly (i.e. in a way that perfectly matches the real world), and estimate the model parameters from high-quality data using estimators that are unbiased and consistent, then with enough data we will recover the true state of nature in our model, which will then be interpretable in a fully causal sense. However, you have likely heard the aphorism "all models are wrong but some are useful" (attributed to the statistician George Box). By "wrong", we mean here that the quantitative details of a statistical model are extremely unlikely to match the true state of nature. This can be the case for many reasons, but a few would be that we have "omitted variables" (variables that play a causal role but are not available to us), or that our model over-simplifies the quantitative relationships among observed variables, e.g. treating effects as linear and additive when they in fact are not.

With sufficient data, semi-parametric and non-parametric modeling methods can be employed, and since such approaches are more flexible, and place fewer restrictions on the assumed model, it is likely that fitted non-parametric models are less likely to be biased (in a way that harms causality) than fitted parametric models, especially parametric models that are very simple. However, non-parametric models generally require more data in order to be fit accurately. As discussed earlier, total error is the sum of systematic error (bias) and random error (imprecision). Thus, in many cases parametric models suffer more from systematic error while non-parametric models suffer more from random error. Since total error is the main threat to the validity of our findings, the choice between flexible non-parametric models and somewhat less flexible parametric models is not always easy to make.

Apart from the models and estimation procedures, the data themselves can sometimes constitute an obstacle to reaching valid causal conclusions. We may have a biased sample (one that doesn't represent the population). While some sampling biases can be statistically corrected, this is usually only the case when the nature of the sampling bias is known. Measurement error, due to factors such as technical noise or even adversarial data corruption can also harm causality, even if our models are perfectly specified.

3.3.4 Targeted methods of causal inference

To overcome the challenges of fully non-parametric analysis while retaining their advantages in terms of bias, a number of techniques have been developed that exploit the fact that in many research studies, only one or a small number of explanatory variables is a priority for causal interpretation, with other factors being secondary. These secondary factors are sometimes called *nuisance factors*. For example, we may have a demographically diverse population of subjects in terms of age, sex, and race, but our interest is only whether on average those subjects who are treated have better outcomes than those who are not treated. To rigorously assess the treatment effect, we need to account for confounding and effect modification by the demographic factors, but all parameters related to the demographic factors are *nuisance parameters* and hence not of primary interest.

Rather than attempting to model all factors non-parametrically to the same extent, it is possible to use more targeted techniques that focus only on establishing the causal role of one particular factor of interest. One early example of this type of approach was *propensity score analysis*. We will not provide a full treatment of this method here, but briefly, it involves modeling the propensity (probability) for a subject to be treated, which is not constant when there is confounding. It is then possible to match, adjust, or weight an analysis that compares the outcome between treated and untreated subjects. Accounting for the propensity scores in one of these three ways can reduce or eliminate the confounding without jointly modeling all relationships among the measured variables.

Other methods for targeted causal inference include g-computation and synthetic controls. Roughly speaking, g-computation uses a model fit to the controls to synthesize a counterfactual control for each of the treated cases. These counterfactual controls match their corresponding subject on all nuisance factors, but are always untreated or unexposed to the risk factor of interest. Some of these methods enjoy a property known as *double robustness*, meaning that valid results are obtained if either the treatment assignment (confounding) relationship is correctly modeled, or if the relationship between the outcome and the treatment is correctly modeled, but not necessarily both.

3.3.5 Brief overview of power analysis

Above we discussed threats to causal validity that arise due to systematic issues with the data or analytic procedures. The other reason that our results can be wrong is due to random errors in our estimated quantities. To control the risk of this happening, most research studies undergo a power analysis before being undertaken.

Considered broadly, power analysis refers to any effort that aims to quantify the likelihood of a data analysis yielding a definitive result, before the data are in-hand. It is important to note that "definitive results" does not mean "positive" or "favorable" results, only that the results are relatively conclusive.

An easy way to get started thinking about power analysis is through the fundamental concepts of *standard errors* and *confidence intervals*. Suppose that we wish to estimate a parameter θ of interest, and we will report the point estimate $\hat{\theta}$ and an approximate 95% confidence interval, which often has the form $\hat{\theta} \pm 2 \cdot \mathrm{SE}(\hat{\theta})$. Your audience will generally have an opinion on how narrow this confidence interval must be in order to be useful, e.g. if $\hat{\theta}$ is a proportion, a useful confidence interval will generally be no wider than 0.1. In a basic analysis focusing on the mean of independent and identically distributed data, the standard error is $\mathrm{SE}(\hat{\theta}) = \sigma/\sqrt{n}$, where σ is the standard deviation of the data and n is the sample size. Even without any data in-hand, it is likely that we have a fairly strong sense of the values of σ, perhaps based on other related data, the scientific literature, or just intuition. If we are not able to speculate on its value, we may be able to provide a reasonable (conservative) upper bound for it. The desired power, made concrete as the width of the 95% confidence interval, is characterized through the equation $4\sigma/\sqrt{n} = w$, where w is the reasonable width for the confidence interval. Solving this for n yields $n = 16\sigma^2/w^2$. This illustrates how the sample size needed to achieve acceptable power can be inferred from assumed information about the target population, when the analysis is focused on point estimation and estimation precision.

The more conventional setting for power analysis is in the context of null hypothesis significance testing (NHST), which is the most common approach to statistical hypothesis testing. This involves stating a null hypothesis and determining a test statistic that measures evidence against the null hypothesis. Commonly, the null hypothesis is a Z-score formed as the ratio of a point estimate to its standard error. Suppose we wish to test the null hypothesis $\beta_j = 0$, where β_j is a certain coefficient in a regression model. We can use the Z-score $Z_j = \hat{\beta}_j/\mathrm{SE}(\hat{\beta}_j)$ to test this hypothesis. For large enough samples, the sampling distribution of Z_j is approximately standard normal, while for small samples it is more appropriate to use a t-distribution with the appropriate degrees of freedom.

While widely available statistical software provides bespoke methods for obtaining power in a variety of specialized settings, here we discuss an approach that can be carried out without special code, and is widely applicable. The derivation is slightly technical, but the result is intuitive and easy to remember.

The power for a formal hypothesis test is the probability of rejecting the null hypothesis when the alternative hypothesis is true. Suppose we wish to achieve 80% power (a common but arbitrary convention) to detect a given effect size (we will discuss effect sizes more below). For the common setting where type-1 error is to be controlled at the 5% level, we reject the null hypothesis when the test statistic is larger than 2. In most settings, the alternative hypothesis is composite, meaning that there are infinitely many alternative hypotheses. Those alternative hypotheses that are very close to the null are almost impossible to detect, while those that are far from the null are easy to detect. This is where the effect size comes in. Calculating probabilities under the alternative hypothesis, we have

$$
\begin{aligned}
P(Z_j > 2) &= P(\hat{\beta}_j/\mathrm{SE}(\hat{\beta}_j) > 2) \\
&= P((\hat{\beta}_j - \beta_j + \beta_j)/\mathrm{SE}(\hat{\beta}_j) > 2) \\
&= P(Z > 2 - \beta_j/\mathrm{SE}(\hat{\beta}_j)) \\
&= P(Z > 2 - \beta_j\sqrt{n}/s)
\end{aligned}
$$

where Z is standard normal, and we write $\mathrm{SE}(\hat{\beta}_j) = s/\sqrt{n}$ (in the case of ordinary least squares, $s = \sqrt{\mathrm{Var}(Y|X)\mathrm{VIF}_j/\mathrm{Var}(X_j)}$). The power is now given by

$$P(Z > 2 - e\sqrt{n}),$$

where $e = \beta_j/s$. We will discuss the interpretation of e below.

Recalling that we seek 80% power, under normality of Z we would need $2 - e\sqrt{n} = -0.84$ (the constant -0.84 is the 20^{th} percentile of the standard normal distribution). This can be rearranged as $2.84 = e\sqrt{n}$. We can now present this relationship in two different ways, for two different purposes. One approach solves for e yielding $e = 2.84/\sqrt{n}$. This gives us the detectable effect size in terms of the sample size. Smaller detectable effect sizes are better, and we can see that the detectable effect size shrinks with n at rate $1/\sqrt{n}$.

Now we consider further how to interpret the effect size e in the context of ordinary least squares. We can rewrite it as $e = \beta_j \cdot \mathrm{SD}(X_j)/(\mathrm{SD}(Y|X) \cdot \mathrm{VIF}_j^{1/2}) = \tilde{\beta}_j/\mathrm{VIF}_j^{1/2}$. The factor $\tilde{\beta}_j = \beta_j \cdot \mathrm{SD}(X_j)/\mathrm{SD}(Y|X)$ is a very natural standardized effect size for β_j. It represents the change in the expected outcome when increasing X_j to $X_j + \mathrm{SD}(X_j)$, in units of $\mathrm{SD}(Y|X)$. Thus, both X and Y are interpreted in standardized units. Rearranging one last time, in the case of OLS we have $\tilde{\beta}_j = 2.84 \cdot \mathrm{VIF}_j^{1/2}/\sqrt{n}$.

Interestingly, $\mathrm{VIF}_j^{1/2}/\sqrt{n}$ is the standard error for β_j if we use standardized units. This shows us that the detectable effect size at 80% power is around 3 (rounding 2.84 up to 3) times the standard error. We can reject the null hypothesis if the estimate is at least two times the standard error, but to have 80% power, we need the population parameter to be almost three times the standard error. This should not be surprising, because if the effect size is equal to the rejection threshold (2 in this case), we will only reject 50% of the time, which is lower than our target power of 80%. This logic turns out to be quite broadly applicable. If we have a reasonable estimate (or projection) for the standard error of the estimator of our target parameter, a safe value to use for the detectable effect size is around three times this standard error.

A different way to use this result is to solve the equation $e = 2.84/\sqrt{n}$ for n, which yields $n = 2.84^2 \cdot \mathrm{VIF}_j/\tilde{\beta}_j^2$. Rounding 2.84^2 to 9, we see that the sample size needed to detect a standardized effect of $\tilde{\beta}_j$, when the variance inflation factor is VIF_j, is around $9 \cdot \mathrm{VIF}_j/\tilde{\beta}_j^2$. In practice, realistic effect sizes are usually much less than 1, so, for example, if the target effect size is $\tilde{\beta}_j = 1/2$, we would need a sample size of $n = 36$, even in the ideal setting where the variance inflation factor is 1. To detect an effect size of 1/4, we would need a sample size of 144 (again with VIF equal to 1).

In summary, we have two methods for approaching power analysis in the setting where the primary data analysis will involve a null hypothesis significance test. One approach gives you the effect size that can be detected at a given sample size, and the other gives you the sample size needed to detect a given effect size. The constants determined above (2 and 9) are specific to the setting where the type-1 error is to be controlled at 0.05 and the power is to be at least 80%, but the values of these constants for other levels of error control can easily be obtained.

The approach of determining the detectable effect size for a given sample size would typically be used when the sample size is constrained by external factors, and the goal is to demonstrate that it is worth investing resources into performing the analysis given the amount of accessible data. The approach of determining the sample size that yields power to detect a given effect size would be useful if we intend to collect data, and there is a sense of what effect size might be plausible. In this case, we can determine how large of a sample would be needed to detect the plausible effect size, and then we would need to decide if it is practical to obtain a sample of the given size.

3.3.6 Power assessment for quantile regression analyses

This is a more advanced topic and is provided for those who are interested. Suppose we wish to conduct an analysis using quantile regression, at a given probability point p. How can we assess the detectable effect size for our planned analysis, or determine the sample size needed to detect an effect of a given size? The situation is more difficult than in the case of ordinary least squares, but is still tractable. For quantile regression targeting the p^{th} quantile of the distribution of Y given X, the covariance matrix of $\hat{\beta}$ can be approximated as $p(1-p)S^{-1}HS^{-1}/n$, where $H = \text{cov}(x)$ is readily estimated, and S is the so-called quantile density matrix, which is defined as $E[f_u(0|x)xx^T]$, where f_u is the density function of $y - Q_p(y|x)$. This can be interpreted as a weighted covariance matrix of x, weighted by the quantile density f_u. For extreme quantiles, the center of the distribution of the quantile residuals $y - Q_p(y|x)$ will be shifted far from zero, hence the density of quantile residuals at 0 will be small. Since the quantile density matrix is to be inverted, this inflates the standard errors, especially for extreme quantiles.

In a power analysis setting, it is often possible to posit a reasonable value for H, but it is more difficult to do so for S. Under Gaussianity, we could set $S = \sigma^{-2} \cdot d \cdot H$, where $d = \exp(-y_p^2/2)/\sqrt{2\pi}$, the standard normal density function evaluated at its p^{th} quantile y_p and σ is the standard deviation of the residuals. Other (non-Gaussian) distributions could be used if heavier tails are anticipated. This calculation also assumes a location shift model in which $y - Q_p(y|x)$ is independent of x. This assumption seems practically unavoidable in the power assessment, but the analysis itself would remain robust to violations of this condition.

The resulting expression for $\text{cov}(\hat{\beta})$ in a quantile regression analysis is $p(1-p)\sigma^2 H^{-1}/(nd^2)$, which can be estimated as $p(1-p)\sigma^2(X^TX)^{-1}/d^2$. This is similar to the case of ordinary least squares, which has covariance matrix $\sigma^2(X^TX)^{-1}$.

When $p = 1/2$ (median regression) and the regression errors are standard Gaussian, the covariance matrix for $\hat{\beta}$ is $2\pi/4 \approx 1.6$ times what we would have for OLS (so the standard errors are inflated by approximately 1.25). This is the cost of using quantile regression to estimate the median – while the conditional median is equal to the conditional mean for Gaussian errors, per the Gauss-Markov theorem, OLS achieves the minimum variance (at least for unbiased estimators), so it is anticipated that a non-least squares approach, while more robust, will be somewhat less efficient.

3.3.7 Assessment Instrument

- Conduct a minimal power assessment that supports the analytic plan you wrote in Section **??**. You may consider a simplified version of the analytic plan to make the power analysis more straightforward. Your assessment should include: the target parameter(s) and the effect size(s) you aim to detect; the assumed or estimated values for key quantities (e.g., residual variance, sample size, VIF); and the resulting detectable effect size or required sample size, with an interpretation of whether the available data are adequate for your aims.

- Write a short memo to yourself, no more than half a page in length, that summarizes the results of your power analysis, as well as any other reasons unrelated to power that the findings resulting from implementing your analytic plan may be misleading or spurious. Consider: potential sources of bias (e.g., confounding, informative missingness, measurement error); threats to causal interpretation given the observational nature of the data; and any limitations of the statistical methods you have chosen (e.g., model misspecification, sensitivity to distributional assumptions).

3.4 Evaluation Rubric

Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
1. Research Aim Formulation (10%)	Research aim is clearly stated, scientifically grounded, and specific; reflects mechanistic or causal thinking rather than pure prediction; scope is well-defined and feasible given the NCHS data; connects to identified gaps in the current state of knowledge.	Research aim is clearly stated and mostly grounded in the scientific domain; scope is reasonable but could be more precisely defined; some connection to causal or mechanistic thinking is evident.	Research aim is present but vague or overly broad; limited grounding in the scientific domain; framed primarily in terms of prediction or statistical methods rather than scientific questions.	Research aim is missing, poorly articulated, or not addressable with the NCHS data; no evidence of engagement with the scientific domain or causal thinking.
2. Dataset & Study Design Understanding (15%)	Memo demonstrates thorough understanding of the dataset's structure, capacity, and limitations; correctly characterizes the data as observational, administrative, clustered, and partially observed; identifies units of analysis, variable types, and population represented; discusses data quality and missingness concerns with specificity.	Memo addresses most key features of the dataset and study design; correctly identifies most data characteristics; data quality and missingness discussed with minor gaps or imprecision.	Memo acknowledges some dataset features but treatment is incomplete or superficial; several important characteristics (e.g., clustering, informative missingness) overlooked or incorrectly described.	Memo is missing or shows little understanding of the dataset's structure, design, or limitations; key characteristics incorrectly described or absent.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
3. Causal & Mechanistic Thinking (15%)	Correctly identifies plausible exposures, outcomes, confounders, and precision variables; references the causal diagram appropriately; demonstrates nuanced understanding of mediation, moderation, or collider bias as relevant to the research aim; acknowledges limitations of causal inference from observational data.	Identifies key exposures, outcomes, and confounders with reasonable justification; references the causal diagram; understanding of causal concepts is mostly sound with minor gaps.	Some exposures and outcomes identified but confounders or mediators are overlooked or incorrectly classified; causal diagram referenced superficially or not at all; limited engagement with causal reasoning.	Fails to identify appropriate exposures, outcomes, or confounders; no engagement with causal or mechanistic thinking; causal diagram ignored.
4. Analytic Plan: Methods & Justification (20%)	Analytic plan is specific, reproducible, and well-motivated; selects appropriate regression or analytic methods with clear rationale; addresses confounding, effect modification, and/or clustering; identifies outcome, key exposures, and control variables; discusses limitations and assumptions of the chosen approach.	Analytic plan is mostly specific and reproducible; methods are appropriate with reasonable justification; most analytic considerations (confounding, clustering) addressed with minor gaps; limitations acknowledged.	Analytic plan is present but lacks specificity or reproducibility; methods chosen without adequate justification; some analytic considerations addressed superficially; limitations mentioned but not developed.	Analytic plan is missing, vague, or infeasible; methods inappropriate or unjustified; key analytic considerations (confounding, clustering, variance structure) absent; no discussion of limitations.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
5. Descriptive Statistics & Effect Sizes (10%)	Proposes at least one well-chosen descriptive statistic or effect size that provides genuine initial insight; includes both absolute and relative quantifications where appropriate; interprets the effect size in context and connects it to the research aim.	Proposes a reasonable descriptive statistic or effect size; interpretation is mostly clear; one of absolute or relative quantification may be missing or underdeveloped.	A descriptive statistic is mentioned but poorly chosen or inadequately interpreted; connection to the research aim is weak; no distinction between absolute and relative effect sizes.	No descriptive statistics or effect sizes proposed; or proposed statistics are irrelevant, incorrectly computed, or uninterpreted.
6. Power Analysis & Uncertainty (15%)	Power assessment is correctly structured and clearly presented; target parameters and assumed quantities are explicitly stated and justified; detectable effect size or required sample size is calculated and interpreted; conclusion about data adequacy is well-reasoned; demonstrates understanding of the relationship between sample size, effect size, and precision.	Power assessment is mostly correct with reasonable assumptions; key quantities identified; interpretation is sound with minor gaps in justification or precision.	Power assessment is attempted but contains errors in setup or calculation; assumptions are stated but poorly justified; interpretation is superficial or partially incorrect.	Power assessment is missing, fundamentally flawed, or demonstrates misunderstanding of core concepts (e.g., effect size, standard error, sample size relationships).

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
7. Bias, Validity & Limitations (10%)	Memo identifies multiple specific threats to validity beyond power, including at least two of: confounding, informative missingness, measurement error, model misspecification, or threats to causal interpretation; discussion is concrete and tied to the specific research aim and dataset.	Identifies at least two threats to validity with reasonable specificity; discussion is mostly tied to the research context with minor gaps.	Acknowledges that bias or limitations exist but discussion is generic or lists threats without connecting them to the specific study; only one substantive threat identified.	Threats to validity are missing or incorrectly described; no meaningful engagement with potential sources of bias or limitations of the analysis.
8. Overall Quality, Communication & Integration (5%)	All three assessments are coherent and build logically on each other; writing is clear, precise, and uses appropriate terminology from epidemiology and biostatistics; demonstrates thorough command of course concepts across the full unit.	Assessments are mostly coherent and well-integrated; writing is clear with minor lapses in terminology or precision; good command of course concepts throughout.	Assessments are partially coherent; some components feel disconnected; writing is adequate but terminology is sometimes imprecise or inconsistent; course concepts applied unevenly.	Assessments are incoherent or largely incomplete; components do not build on each other; writing is unclear or misuses key terminology; limited evidence of course concept mastery.

Chapter 4

Unit 4: Designing Interpretable Predictive Models

Total Time: 5 hours
Instructional Time: 3.5 hours
Assessment Time: 1.5 hours

This unit introduces the foundations of supervised machine learning models, with a focus on interpretability and communicating modeling decisions. Throughout the unit, the **iBudget Florida Medicaid Waiver Study** (Scientific Report, Volumes I and II, available as reference documents) serves as the primary worked example. The iBudget study evaluated the algorithm that Florida uses to allocate support budgets to over 35,000 individuals with developmental disabilities, building and comparing eight predictive models using real administrative data. It illustrates in a high-stakes, real-world setting every concept this unit covers: baseline model construction, interpretable feature engineering, multicollinearity diagnostics, model comparison, and stakeholder reporting. Volume II, which extends the predictive results into a resource allocation framework, is offered as supplemental reading for awareness; students are not expected to implement its methods. The parallel task throughout this unit is the 2026 Data Challenge, in which students apply the same workflow to NCHS vital statistics data to project underweight births and infant mortality by Texas county.

4.1 Foundations of Predictive Modeling

Time: 1.5 hours (Instructional: 70 minutes, Assessment: 20 minutes)
This lesson introduces supervised learning through the workflow that students will use throughout the rest of the unit: selecting a prediction target, choosing features, fitting an interpretable baseline model, and evaluating it on held-out data. The emphasis is on understanding what the model is doing, not only on obtaining a good score.

4.1.1 Learning Objectives

1. Distinguish supervised learning from other common data analysis tasks, and distinguish regression from classification.

2. Before fitting any model, articulate the prediction problem in research terms: name the outcome variable and how it is operationalized in the dataset, identify the unit of analysis, describe the population represented, and state the intended use of the forecast.

3. Build a linear regression model as an interpretable baseline and explain what its coefficients represent.

4. Practice the workflow for performing a train-test split, training a model on training data, evaluating it on held-out test data, and understanding how cross-validation works in practice.

5. Identify common risks in predictive modeling, including overfitting and data leakage.

4.1.2 Instructional Notes

The iBudget study opens with exactly the framing exercise described in Learning Objective 2. Before any model is built, the study defines its target variable (annualized support cost per enrolled client), its unit of analysis (individual clients in the Florida Home and Community-Based Services waiver), and the operational use of the prediction: the model output becomes the need estimate that informs each client's budget allocation. This framing step is not bookkeeping; it determines what a good model means. A model that predicts well on average but systematically underpredicts costs for the highest-need clients may be statistically respectable and operationally harmful at the same time.

For the data challenge, students face an analogous framing decision. The outcome could be operationalized as the county-level rate of underweight births (a proportion), as the raw count of underweight births, or as a binary indicator of whether a county exceeds a threshold rate. Each operationalization implies a different model family and a different error cost. Students should make this choice explicitly and record it before building their baseline.

The iBudget baseline is the frozen 2015 model: a square-root transformed linear regression with coefficients fixed at their 2015 values, applied to FY2024–25 data without recalibration. This is a meaningful baseline, not a straw man. It answers the question of whether the operational model has drifted from the conditions under which it was calibrated. Students should draw the analogy to their own baseline choice: what is the simplest defensible model, and what question does running it answer?

The 2015 iBudget study reported only in-sample metrics, which the 2025 reevaluation identifies as a critical limitation. This is a concrete illustration of why the train-test split discipline in Learning Objective 4 matters: in-sample fit is not evidence of predictive accuracy, and a model evaluated only on its training data cannot be trusted to generalize.

4.1.3 Assessment Instrument (20 minutes)

Students will build a baseline linear regression model using the Vital Statistics dataset. They will document the outcome variable and explain how it is operationalized in the NCHS records, list the features used, report one test-set performance metric such as RMSE or MAE, and write a brief note identifying two ways data leakage or overfitting could occur in this workflow.

4.2 Interpretable Feature Engineering and Pipelines

Time: 1 hour (Instructional: 40 minutes, Assessment: 20 minutes)
This lesson focuses on interpretable feature engineering: creating features that better reflect patterns in the data while preserving interpretability. Students will use plots and domain knowledge to motivate simple transformations, encodings, and interactions, and will organize these steps within a small `sklearn` Pipeline.

4.2.1 Learning Objectives

1. Use visualizations to spot patterns in the data that may motivate feature engineering, such as nonlinearity, thresholds, and group differences.

2. Understand when to use simple transformations, interaction terms, and common encoding strategies such as one-hot encoding.

3. Build a small preprocessing and modeling Pipeline in `sklearn` and explain how the resulting features affect model interpretability.

4.2.2 Instructional Notes

The iBudget study's feature engineering process illustrates how visualizations motivate transformations. Before any model was fit, the study generated box plots of budget utilization by living setting, by age group, by race and ethnicity, and by county, and a scatter plot of budgeted amount against actual spending. Each of these plots revealed a pattern that motivated a feature decision. Utilization varied substantially by living setting, which motivated including it as a predictor. The budget-versus-spending scatter plot revealed a right-skewed outcome distribution, which motivated considering log and square-root transformations of the response. Students can apply the same discipline: plot the outcome against each candidate feature and let the pattern motivate the engineering decision.

The iBudget study also illustrates a subtler challenge that one-hot encoding alone does not resolve: categorical variables whose levels carry different meanings than their labels suggest. The living setting variable originally had 22 levels, many of which were sparsely populated and clinically overlapping. The study consolidated these to 6 levels through a combination of statistical analysis and domain judgment. One-hot encoding 22 sparse levels would have produced an unstable and uninterpretable set of dummy variables; consolidating first, then encoding, produced a feature that was both informative and legible. Students should look for analogous situations in the NCHS data, particularly in variables with many low-frequency categories.

The `sklearn` Pipeline is the right tool for keeping these transformations organized and reproducible. A Pipeline ensures that preprocessing steps applied during training are applied identically to test data, which prevents a common form of data leakage.

4.2.3 Assessment Instrument (20 minutes)

Students will continue working with the Vital Statistics dataset and will add two or three engineered features in a small `sklearn` Pipeline. They will submit the pipeline structure and a short written justification for each added feature, including why it may improve model fit or interpretability.

4.3 Feature Justification, Multicollinearity, and Model Diagnostics

Time: 1.5 hours (Instructional: 60 minutes, Assessment: 30 minutes)
Building upon Section **??**, students will learn how to decide whether a feature should remain in a model. The emphasis is on practical diagnostics and justification using correlations, multicollinearity checks, variance inflation factors, coefficient stability, and residual plots.

4.3.1 Learning Objectives

1. Use simple quantitative tools such as Pearson and Spearman correlation, multiple R^2, and variance inflation factor to identify useful, redundant, or unstable features and to recognize multicollinearity.

2. Use residual plots and side-by-side model comparison to assess whether added complexity is justified.

3. Explain why a feature was kept, transformed, or removed based on predictive performance, interpretability, and stability.

4. Examine model performance separately for meaningful subgroups, such as geographic region or demographic category, and recognize when a model that performs well overall may perform unevenly across the population it serves.

4.3.2 Instructional Notes

Multicollinearity is not a pathology to be feared; it is information about the feature space that must be understood and managed. The iBudget study screened candidate features using mutual information with permutation baseline correction across five consecutive fiscal years, then checked pairwise redundancy using Spearman rank correlation for continuous-continuous pairs and bias-corrected Cramér's V for categorical pairs. VIF screening followed for the final candidate set. This sequence is exactly the workflow Learning Objective 1 describes; students can treat Chapter 5 of the iBudget Volume I as a methodological reference.

The iBudget study also illustrates why comparing more than two models is scientifically honest. The study evaluated eight models under a common protocol: an 80/20 train-test split with 10-fold cross-validation, all metrics computed on the held-out test set. The comparison included OLS re-estimation, a Generalized Linear Model with a Gamma distribution (which handles the positive, right-skewed distribution of support costs more appropriately than OLS), robust regression using Huber M-estimation, ridge regression to manage multicollinearity among correlated assessment items, and a random forest as the accuracy ceiling. Students do not need to implement all of these, but they should understand why a family of models exists: each alternative addresses a specific limitation of the OLS baseline that can be diagnosed from residual plots and distributional checks. A fitted-vs-residual plot that shows a funnel pattern suggests heteroscedasticity; a GLM or weighted least squares may be warranted. A pattern of large positive residuals for the highest-cost observations suggests the outcome distribution needs a transformation or a different family entirely.

Learning Objective 4 reflects a lesson that emerges clearly from the iBudget subgroup analysis. A model that predicts overall support costs accurately can still systematically underpredict costs for the highest-need clients, whose budgets have the largest consequences for service access. In the data challenge context, a county-level model that performs well on average across Texas may perform poorly on rural or border counties, which are precisely the populations where targeted public health intervention is most important. Subgroup performance is not optional reporting; it is part of what the model works means.

4.3.3 Assessment Instrument (30 minutes)

Students will compare two candidate models, such as a baseline model and an expanded feature-engineered model. They will decide which model they would keep and justify the decision using at least three pieces of evidence drawn from performance metrics, multicollinearity checks, coefficient stability, or residual diagnostics. The justification should also include one observation about whether the preferred model's performance is consistent across at least one meaningful subgroup in the data.

4.4 Model Reporting and Stakeholder Communication

Time: 1 hour (Instructional: 30 minutes, Assessment: 30 minutes)
This lesson focuses on communicating model performance and design decisions to external stakeholders. The goal is to turn the analytic work from Section **??** into a clear written summary that reports model results responsibly and in a form that aligns with the broader reporting expectations introduced earlier in the curriculum.

4.4.1 Learning Objectives

1. Understand how to report and compare interpretable predictive models using appropriate regression metrics such as MSE, RMSE, and MAE.

2. Summarize model design choices, strengths, and limitations in language appropriate for a non-technical audience.

3. Practice connecting model reporting to broader ideas of transparency and reproducibility, including the spirit of TRIPOD-style reporting.

4.4.2 Instructional Notes

The iBudget study's reporting structure is a professional example of the TRIPOD spirit applied to an administrative prediction problem. Each model in the comparison follows an identical chapter template: specification, overall performance metrics, accuracy bands, cross-validation results, subgroup analysis, variance and stability metrics, and implementation considerations. This structure means that any reader can navigate to any modeling decision and find it documented. Nothing is left to be inferred from code. Students do not need to produce a report of that scope, but they can use it as a reference for what complete documentation looks like and what questions a careful reader will ask.

The report produced in this section will be carried forward. Unit 6 asks whether estimates are stable across data sources and time periods, and a model report that does not specify its outcome operationalization or validation strategy cannot be meaningfully compared to estimates from other sources. Unit 7 subjects the report to LLM ensemble critique, and a report with gaps will produce ensemble outputs that are difficult to verify. Writing the report well here has direct consequences for the quality of work in both subsequent units.

4.4.3 Assessment Instrument (30 minutes)

Students will write a short report for a stakeholder describing their final model. The report should identify the prediction target, summarize the features included, report one or two performance metrics, and explain at least one important limitation or caveat in plain language. The report should also include one sentence connecting the model to the data challenge: what specific projection will this model contribute to the final analysis, and for which population or geography?

4.5 Evaluation Rubric

Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
1. Foundations of Predictive Modeling (25%)	Clearly distinguishes supervised vs. unsupervised, regression vs. classification; correctly performs a train-test split, trains an interpretable baseline model, evaluates it on held-out data, and explains how cross-validation works; articulates the prediction problem in research terms before fitting; demonstrates strong understanding of overfitting and data leakage.	Good workflow with minor errors in reasoning or terminology; prediction framing present but partially developed; overfitting/data leakage described but not deeply.	Workflow partially correct; inconsistent understanding of splits, evaluation metrics, or model fitting; prediction framing absent or generic.	Incorrect or incomplete workflow; poor understanding of model evaluation, overfitting, or data leakage; no research framing of the prediction problem.
2. Interpretable Feature Engineering & Pipelines (20%)	Insightfully uses visualizations to identify meaningful patterns; appropriately applies simple transformations, encoding strategies, and interaction terms; builds a clear `sklearn` Pipeline; thoroughly documents the rationale for each feature added.	Good feature construction with minor justification gaps; Pipeline mostly correct; documentation adequate.	Basic feature creation with limited insight; Pipeline incomplete or justification superficial.	Poor or incorrect feature construction; missing Pipeline; unclear or unjustified decisions.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
3. Feature Justification, Multicollinearity & Diagnostics (25%)	Correctly applies feature justification tools such as correlation checks, multiple R^2, variance inflation factor, and residual diagnostics; correctly identifies multicollinearity concerns; compares models thoughtfully and gives a clear, evidence-based rationale for keeping the preferred model; includes at least one subgroup performance observation with genuine analytical content.	Most diagnostics applied correctly; model comparison reasonable but with minor gaps in interpretation or justification; subgroup observation present with adequate reasoning.	Partial or inconsistent use of diagnostics; model choice weakly justified; subgroup observation absent or superficial.	Incorrect or missing diagnostic evidence; model choice unsupported or inconsistent with results; no subgroup analysis.
4. Model Reporting & Stakeholder Communication (20%)	Accurately computes and interprets regression metrics such as MSE, RMSE, and MAE; report is clear, well-structured, and communicates model design decisions, performance, and limitations appropriately for the intended audience; includes a specific, meaningful connection to the data challenge.	Metrics computed correctly with minor errors; report mostly clear and appropriately framed; data challenge connection present but general.	Metrics included but inconsistently computed or interpreted; explanation limited or too technical; data challenge connection absent or formulaic.	Metrics missing or incorrect; report unclear, incomplete, or not aligned with results; no connection to the data challenge.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
5. Overall Professionalism, Clarity, and Documentation (10%)	Writing polished and well-organized; notebooks/scripts clean, reproducible, and well-commented; outputs easy to interpret.	Mostly clear with minor structural or formatting issues; code readable.	Understandable but inconsistent structure or documentation.	Poorly organized, unclear writing, missing documentation, or sloppy code.

Chapter 5

Unit 5: Reproducible Workflows

Total Time: 5.5 hours

5.1 Goals of Reproducible Analyses

Learn the key goals and challenges of creating reproducible, transparent, and user-friendly analyses that are easy to share and reuse.

5.1.1 Learning Objectives

1. Awareness of key challenges and goals when creating reproducible workflows, including making analyses reproducible, user friendly, transparent, reusable, version controlled, and archived.

5.1.2 Assessment Instrument

1. Write in your own words a summary of the key challenges and goals when creating reproducible workflows as it applies to data generally, your own work, and the CDC dataset.

5.2 Reproducibility via Code Notebooks

Gain awareness of Markdown, Jupyter, and Quarto, and learn how these tools integrate to create clear, reproducible workflows for data analysis and reporting.

5.2.1 Learning Objectives

1. Awareness of Markdown, Jupyter, Quarto, and how these tools can be integrated into reproducible workflows.

5.2.2 Assessment Instrument

1. Create a notebook that does some simple EDA on some data. It should generate at least one plot and at least one table. Create a script to download the CDC dataset. Upload your work.

5.3 Best Practices for Reproducible Programming

Learn essential best practices for reproducible programming, including writing clear scripts and functions, avoiding magic numbers, using caching and seeding for randomness, and refactoring code to enhance clarity, reliability, and repeatability.

5.3.1 Learning Objectives

1. Awareness of best practices for reproducible programming including writing scripts, functions, avoiding magic numbers, caching and seeding randomness, and how to refactor code to align with these practices.

5.3.2 Assessment Instrument

1. Add documentation to your simple EDA notebook. Create a Makefile similar to the one shown in class (e.g., with commands to download the data, run the analysis, etc.). Upload your analysis file.

5.4 Version Control

Gain a basic understanding of Git, its advantages, and learn to perform essential tasks such as cloning repositories, committing changes, and syncing with remote repositories using push and pull commands.

5.4.1 Learning Objectives

1. Familiarity with Git and its benefits, and the ability to begin using it for simple tasks, including cloning, committing changes, pushing and pulling.

5.4.2 Assessment Instrument

1. Put the source code for a template project on GitHub. Submit the link to your GitHub page. Describe any challenges you encounter.

5.5 Containers

Gain hands-on experience with key dependency management tools (Python virtual environments, renv, and containerization), understanding their pros and cons, and develop the skills to create and run basic Docker images.

5.5.1 Learning Objectives

1. Familiarity with various tools for dependency management, including Python virtual environments, renv, and containerization, and their respective strengths and weaknesses. Ability to create and run simple Docker images.

5.5.2 Assessment Instrument

1. Put a template project into an image, and ensure that it runs as expected. Upload the Dockerfile/Containerfile and other source files to the GitHub project. Describe any challenges you encounter.

5.6 Assembling a Full Analysis Pipeline

Learn key factors in organizing an analysis pipeline and develop the skills to assemble a complete, reusable pipeline template.

5.6.1 Learning Objectives

1. Considerations when organizing an analysis pipeline, and the ability to assemble a full template pipeline.

5.6.2 Assessment Instrument

1. Create a template project. It should have a directory structure similar to the one shown in class (e.g., with a `data` directory, `analysis` directory, etc.) and include a few simple markdown files. Push the full analysis to GitHub. Describe any challenges you encounter.

5.7 Evaluation Rubric

Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
1. Goals of Reproducible Analyses (10%)	Provides a thorough summary, clearly articulating key goals and challenges of reproducible workflows; connects concepts to personal work and the course dataset; demonstrates careful reflection on transparency, usability, version control, and archiving.	Addresses most key challenges and goals clearly; provides some connection to personal work and dataset; demonstrates solid understanding with minor omissions.	Identifies basic challenges and goals, but with limited depth or reflection; may not connect well to examples or may skip some elements.	Incomplete or superficial summary; lacks understanding of core concepts or fails to address relevance to own work.
2. Reproducibility via Code Notebooks (20%)	Creates a well-structured notebook with reproducible code, clear markdown, at least one plot and one table; documentation is clear; script downloads CDC dataset; all work uploaded; demonstrates integration of tools.	Notebook includes most required elements (code, markdown, plot/table); documentation is mostly clear; dataset obtained; minor gaps in reproducibility or clarity.	Notebook has basic elements, but some required components (plot/table, markdown, or script) are missing or poorly integrated; minimal documentation.	Notebook missing key requirements; poorly documented or not reproducible; files not uploaded, or CDC dataset not included.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
3. Best Practices for Reproducible Programming (15%)	Adds thorough documentation to notebook; creates a well-organized Makefile with clear, reusable commands (e.g., download, run analysis); analysis file is uploaded; demonstrates strong understanding of reproducible scripting (functions, no magic numbers, caching, seeding randomness, refactoring).	Documentation and Makefile present and mostly clear; analysis file uploaded; most best practices followed, though minor omissions may be present.	Documentation or Makefile is incomplete or unclear; basic understanding of practices is evident, but with gaps in execution or clarity.	Documentation and/or Makefile missing or severely lacking; little evidence of understanding reproducible programming practices.
4. Version Control (15%)	Template project code on GitHub is complete and well-structured; the link is shared; meaningful commit history demonstrates staged progress and clarity; challenges are described in detail.	Project is on GitHub with most files complete; link is shared; some commit history and brief discussion of challenges; mostly correct use of Git.	Project present but incomplete or poorly organized; minimal commit history; challenges noted superficially.	Project not on GitHub or files missing; little or no evidence of version control understanding; challenges not described.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
5. Containers & Dependency Management (20%)	Template project is successfully containerized (Docker/renv/venv); runs as expected; Dockerfile and other source files uploaded to GitHub; demonstrates clear understanding of strengths and weaknesses of dependency management tools; challenges discussed in detail.	Container image runs with minor issues; most files uploaded; reasonable explanation of tool choices; challenges discussed briefly.	Container created but may not run as expected, or files/documentation are incomplete; basic understanding of tools, but lacks depth.	Container not created or does not run; missing files or documentation; little or no understanding of dependency management.
6. Assembling a Full Analysis Pipeline (15%)	Creates a comprehensive template project with a clear directory structure (e.g., `data`, `analysis` directories) and markdown documentation; pipeline is pushed to GitHub; demonstrates thoughtful organization, reproducibility, and ease of use; challenges discussed in detail.	Project structure and markdown are mostly clear; pushed to GitHub; most requirements met, with minor organizational or documentation gaps; challenges briefly noted.	Basic directory structure and markdown present; pipeline missing some components or documentation unclear; pushed to GitHub but meets only minimal requirements; challenges superficially addressed.	Template project missing core structure or documentation; not pushed to GitHub or requirements not met; challenges not described.
7. Professionalism, Clarity, Structure (5%)	Work is polished, well-organized, clear, and follows all instructions; files are named and formatted consistently; writing is succinct and logical.	Work is mostly clear and organized; minor formatting or organizational issues.	Basic clarity but contains organizational weaknesses, unclear writing, or some failures to follow instructions.	Work is disorganized, unclear, or does not follow instructions; formatting and naming are inconsistent or missing.

Chapter 6

Unit 6: Data Integration and Meta-analysis

Total Time: 3 hours

6.1 Key Concepts in Data Integration

Time: 1 hour

This lesson provides an introduction to meta-analysis as a tool for quantitative data integration and research synthesis. *Data integration* arises when either raw data or summary statistics are available for many subpopulations, for the same or a similar set of outcomes and predictors. In this setting, the goal is to understand the common and distinct ways that the predictors and outcomes are distributed and are related to each other across the subpopulations. *Research synthesis* arises when results from a collection of research studies considering similar or identical research questions are available. The studies may differ in the populations assessed, in the study designs, or in the methods used. Combining evidence across diverse studies can yield a more accurate *consensus estimate* of the relationships of interest and can reveal whether there is *heterogeneity* among the subpopulations in how these relationships are structured.

This document focuses on design, analysis, and interpretation. It includes no code or numerical results. This Python notebook provides some model analyses from which you can borrow ideas as you start to implement analyses informed by the principles covered here.

6.1.1 Learning Objectives

1. Identify settings where data integration is possible and likely to be informative.

2. Explain the principle of *inverse variance weighting* and why it is more effective than simple averaging.

3. Explain the difference between *uncertainty* and *estimation imprecision* in individual studies or subpopulation-specific results, and contrast this type of variation with heterogeneity in the true values of a parameter in different studies or subpopulations.

4. Explain and contrast error control via the *false positive rate*, the *family-wise error rate*, and the *false discovery rate*, and explain the key ideas behind the local false discovery rate approach.

5. Explain the basic mathematics behind methods for pooling estimates, pooling standard errors, and combining p-values from independent sources, and identify methods for combining p-values that are robust to some forms of dependence.

6. Explain the potential uses of some of the emerging approaches for data integration and assessment of heterogeneity based on *E-values* and *empirical likelihood*.

6.1.2 Introduction to Key Concepts

Although data integration and meta-analysis have many distinct features, at a conceptual level they also share important similarities. We will first focus on these similarities. As is usual in empirical research, we have some notion of a *population* that we wish to study and a parameter of interest θ, such as mean (expected) birth weight. When the population is large and complex, there are usually ways to partition it into more homogeneous subpopulations. While we typically think of a parameter such as mean birth weight as having a single fixed value, upon partitioning the population, the value θ of the parameter may now be specific to each subpopulation indexed by i, and we denote it by θ_i. We can now consider the distribution of θ_i across the subpopulations $i = 1, \ldots, m$, where m is the number of subpopulations.

In the setting of data integration, we often have complete data for all individuals, including their value for the outcome being studied (e.g. birth weight) and the identity of the subpopulation to which they belong. Alternatively, we may only have aggregated data, such as the sample mean birth weight per subpopulation, hopefully accompanied by the number of observations and sample standard deviation for each subpopulation (the reason that this ancillary information is important will become clear later). The latter situation may arise if public individual-level data are not accessible due to privacy concerns. In the NCHS birth data, complete individual-level data for years up to 1988 are available, but we can aggregate it to emulate the situation where only summary data are available.

Meta-analysis usually refers to the setting where we are trying to learn about a population by integrating information from many research studies that aimed to study the same population. Almost inevitably, these research studies will effectively target different subpopulations of the overall population of interest, either intentionally or inadvertently. For example, many researchers draw their human subjects from the local population, so each study will be set in a distinct demographic and geographic context. Moreover, studies almost always have inclusion criteria that define the population being studied, e.g. if one study excludes children and another does not, and these inclusion criteria are rarely consistent across studies.

In a meta-analysis, we usually rely on summarized results reported in the scientific literature. It is almost inevitable that different methods will be used in different published studies. These methodological choices may influence the population being studied (e.g. through differing inclusion criteria), or reflect how parameters are defined and estimated (e.g. the use of different control variables in a regression or different statistical estimators). Studies may also differ in terms of *measurement*, e.g. what instruments are used for data collection.

There are many similarities between data integration and meta-analysis, so similar theoretical and practical tools can be utilized in both settings. It is not necessary to draw a bright line between these two concepts – some situations will be hard to place uniquely under one heading or the other. For example, some people may speak of *complete data meta-analysis*, where one aims to do a research synthesis in a setting where the raw data for all studies are available. In this case, one has the choice to proceed as in a data integration, or as in a meta-analysis, or by adopting some hybridization of the two.

At a high level, there are two main analytic goals that we discuss here. One important goal is to leverage the increased statistical power that results from combining information in many related datasets, or in many distinct publications. This will provide a more accurate estimate of the population parameter of interest. Even if we believe that the underlying parameter of interest varies among the subpopulations, the focus in this case is not on this variation, rather it is on the consensus value that can be obtained by pooling evidence across many sources. The other major analytic goal that we will explore here takes essentially the opposite perspective – that the variation in the parameter across subpopulations, or across methodologies, is of direct interest, and we wish to quantify and interpret the extent to which such heterogeneity is present.

6.1.3 Pooling Evidence from Tests

One common setting for data integration occurs when statistical tests of the same null hypothesis are carried out within many subpopulations. A common analytic goal could be to assess the *global null hypothesis* asserting that the null hypothesis is true in every subpopulation. This task might be best referred to as *evidence integration*, since the goal is to pool the evidence obtained from different subpopulations, instead of pooling the data itself. It can be seen as a modern generalization of the frameworks of *Fisherian* p-values and *Neyman-Pearson* decision theory for null hypothesis significance testing.

In the most conventional setting, we have a properly calibrated p-value p_i for each subpopulation. Here by "properly calibrated" we mean that p_i follows a uniform distribution on $[0, 1]$ if the null hypothesis is true for subpopulation i. The results presented here would also apply if p_i *stochastically dominates* a uniform distribution, meaning that $P(p_i > t) \geq 1-t$ for all t. There are a number of so-called *combining rules* that allow us to assess evidence against the global null hypothesis using either a collection of p-values or a collection of numerical test statistics, one for each subpopulation.

If the tests, and hence the p-values, are statistically independent, then the classical *Fisher combination test* can be applied. This approach proceeds by transforming each of the m p-values to have an exponential distribution when the null hypothesis is true, taking advantage of the fact that $-\log(p)$ follows an exponential distribution with mean 1 when p is uniform on $[0,1]$. Since the sum of independent exponentially distributed random variables follows a χ^2 distribution, we obtain the test statistic $F = -2\sum_{i=1}^m \log(p_i)$, which follows a χ^2 distribution with $2m$ degrees of freedom when the global null hypothesis is true. When the p-values are not independent the second step fails, and the χ^2 distribution is no longer the correct reference distribution.

In recent years, interest has centered on the case where the tests (and hence the p-values) may not be statistically independent. One simple but ineffective approach is to use the largest p-value (the weakest evidence) as a summary of all the p-values – this is a valid statistic (it stochastically dominates the uniform distribution on $[0,1]$) but is very conservative. This led to the introduction of *combining rules* that produce valid p-values under dependence, which was studied here. A natural source of combining rules are the classical "Pythagorean means". It turns out that (i) the arithmetic mean of p-values multiplied by 2 is a valid p-value, (ii) the geometric mean of p-values multiplied by e is a valid p-value, and (iii) the harmonic mean of p-values multiplied by $\log(m)$ is a p-value.

The statistical power of these procedures depends on the nature of the alternative hypotheses. For example, since the harmonic mean is most sensitive to the presence of one or a few very small values, it will be most powerful in a sparse situation where most of the null hypotheses are approximately true, with a few of the subpopulations presenting strong evidence against the null (hence yielding small p-values). In contrast, the arithmetic mean would be most powerful in certain "dense" situations where the evidence against the null is of a similar level in all subpopulations. Of course the structure of the alternative hypotheses is not known ahead of time, and the test to be used must be selected before seeing the data – it is invalid to try multiple approaches and select the one that gives the smallest p-value. Nevertheless there are situations where there is good reason to suspect ahead of time that the alternative hypothesis is either sparse or dense, allowing one to select an *a priori* approach accordingly.

Another class of approaches for pooling p-values begins by transforming them to a given probability distribution represented by a cumulative distribution function (CDF) F. If U is uniformly distributed on $[0,1]$, then $F^{-1}(U)$ is a random variable whose CDF is F – this is known as the "CDF transformation method". After transforming the p-values in this way, they are averaged, and then back-transformed to the uniform scale. This approach proves to be most effective when the distribution F belongs to a family that is closed under summation, and that consists of heavy-tailed distributions. For example, the sum of Cauchy random variables also follows a Cauchy distribution, hence $m^{-1}\sum_i \tan(\pi(1/2-p_i))$ follows a Cauchy distribution. This yields the Cauchy combination test, and a related approach involving the Pareto distribution is known as the Pareto combination test. Transforming the distributions to have heavy tails mitigates most dependence that may exist among the p-values.

6.1.4 Evidence Pooling in Large Scale Inference

In the last 30 years, major advances in methodology for integrating disparate forms of statistical evidence have been made. A subfield of statistics termed large scale inference has developed around these new methods. The new techniques in this area are motivated by modern applications made possible by new ways of taking measurements, with genomic and biological imaging applications being most prototypical, although applications exist in many other areas. The common theme is that evidence is collected about a large number of parameters, and while the total volume of data is large, the information about any particular parameter under study is small. It turns out that ideas from *empirical Bayes* methodology can play a very useful role.

We will discuss one method from large scale inference here, known as the *local false discovery rate*, or local FDR, introduced here. At a surface level, the local FDR is a refinement of an earlier approach to bounding false discovery rates known simply as the *false discovery rate approach*, usually attributed to Benjamini and Hochberg or to Storey. We briefly introduce those earlier ideas here first. Suppose we are testing a large number, say m of hypotheses, and we have statistics T_i, $i = 1, \ldots, m$, such that the larger the value of T_i, the stronger the evidence against the i^{th} null hypothesis. Assume that all the test statistics are on the same scale, so that it makes sense to employ a single threshold t to decide when to reject each null hypothesis. We reject the i^{th} null hypothesis if $T_i > t$.

As always, there is a tradeoff in that greater values of t will yield more specificity and lower sensitivity, and lesser values of t will yield more sensitivity and greater specificity. In the setting of classical multiple hypothesis testing, we might aim to control the *family-wise error rate* (FWER), which would involve setting t so that the under the global null hypothesis (where all null hypotheses are true), the probability of any T_i exceeding t (the FWER) is controlled, say at $\alpha = 0.05$. A practical way to do this that does not impose any constraints on the dependence between the tests is the *Bonferroni approach* (also known as the union bound approach), which sets t sufficiently large so that $P(T_i > t) < \alpha/m$.

To define the false discovery rate, let $H_i = 1$ if the i^{th} null hypothesis is true. The FDR is $E[\sum_i \mathcal{I}(T_i > t) \cdot \mathcal{I}(H_i = 1)/\sum_i \mathcal{I}(T_i > t)]$. The key observation here is that the FDR looks at the false rejection rate among all rejected null hypotheses, not among all tests (which is what FWER controls). A test that is more permissive will have a larger numerator for the FDR but will also have a larger denominator, at least in settings where many of the null hypotheses are false, so that the ratio may remain small. In practice, the FDR is useful when the costs of false positives and false negatives are relatively equal, and it is desirable to make a large number of claims that are “mostly right”. On the other hand, FWER control is desirable when even a single false positive has severe consequences.

The local FDR is related to the traditional FDR, but provides a different perspective based on *empirical Bayes* thinking. Local FDR is usually approached using Z-scores of the form $Z_i = \hat{\theta}_i/\widehat{\mathrm{SE}}_i$. Suppose we have a large number of such Z-scores, some of which correspond to true null hypotheses. The distribution of the "null" Z-scores should follow a standard normal distribution. The non-null Z-scores, if any, would follow a distribution F whose structure would depend on the effect sizes. We do not know which Z-scores are null and which are not, so we observe a mixture distribution $\pi N(0,1) + (1-\pi)F$, where $\pi \in [0,1]$ and F are both unknown.

The local FDR focuses on densities, whereas the traditional FDR focuses on cumulative probabilities (this is what makes it "local"). The local FDR considers the ratio $\pi \cdot f_0/f$, where f_0 is the null distribution and f is the density of all test statistics. We usually know f_0 from the theory governing our Z-scores, e.g. in many cases f_0 will be standard normal. The density function f can be empirically estimated from the observed test statistics. The value of π is difficult to estimate, and in cases where it is anticipated that most null hypotheses are true, it can be replaced with its upper bound which is 1. If we use the simplified form $f_0(z)/\hat{f}(z)$ to estimate local FDR value at z, we are simply asking whether near a particular Z-score z, is the density of observed Z-scores large relative to what would be expected under the global null hypothesis?

As a practical matter, densities are difficult to estimate, especially in the tails of the distribution, which is exactly where we are operating here. In the standard local FDR approach, a method known as Lindsey's method is used to obtain a stable estimate of $\hat{f}$.

A variant of the local FDR procedure uses a data-driven version of f_0 known as the *empirical null* instead of the theory-driven approach, which is usually based on large sample theory. This allows us to use the data to inform us of the behavior of the test statistic under the null, possibly providing a more robust analysis.

6.1.5 Pooling Estimates

Classical tests of significance based on the Neyman-Pearson framework continue to play an important role in modern statistics, but there is a broad consensus that they are over-used and fail to address some important points. A test statistic or test decision tells you something about whether a particular null hypothesis is consistent with the data. One concern about hypothesis testing approaches is that if the null hypothesis is not consistent with the data, i.e. if your decision is to reject the null hypothesis, you are provided little indication of what hypothesis actually is consistent with the data. This may lead to people seeking to reject "straw man" null hypotheses, the falsity of which provides little insight into the true nature of the population.

Unlike a p-value, a parameter estimate provides insight into the direction and magnitude of an effect, and when combined with a standard error can be used for assessing uncertainty and statistical significance. In many cases a parameter estimate along with a statement of precision about the estimate is more informative than evidence against a null hypothesis in the form of a p-value. Here we will discuss some approaches that can be used to integrate effect estimates to achieve this goal.

Consider first the fundamental goal in data integration in which we take a large collection of point estimates, and consolidate them into a single point estimate that provides a consensus estimate for the data represented by the overall population. We will discuss heterogeneity later, but we note now that this consensus estimate could be interpreted as the parameter that holds for all observations in the overall population if there is no heterogeneity, or it could be interpreted as the marginalized version of a collection of heterogeneous subpopulation-specific effects.

6.1.5.1 Pooling Estimates by Averaging

One somewhat naive way to form a consensus estimate is by averaging the point estimates. Let $\hat{\theta}_{\text{avg}} = (\hat{\theta}_1 + \cdots + \hat{\theta}_m)/m$ denote this consensus estimate. Its standard error can be written

$$\mathrm{SE}(\hat{\theta}_a) = \sqrt{m^{-1}\sum_j \mathrm{SE}_j^2}/\sqrt{m}.$$

The numerator of this expression is just the square root of the average of the sampling variances. The denominator $\sqrt{m}$ reflects the fact that we gain precision as we pool a larger number of point estimates.

The pooled estimate $\hat{\theta}_a$ has some good properties, but it is not *efficient*, that is, it does not make full use of the information in the data. Inverse variance weighting allows us to obtain a fully efficient pooled estimate. The theoretically optimal weight for the i^{th} point estimate is $w_i \propto 1/\mathrm{SE}_i^2$. This assigns lower weight to the point estimates with greater variance. The inverse variance weighted point estimate is $\hat{\theta}_{\text{ivw}} = \sum_i w_i\hat{\theta}_i/\sum_i w_i$. The standard error of this estimator is $H^{1/2}/\sqrt{m}$, where $H = m/(1/\mathrm{SE}_1^2 + \cdots 1/\mathrm{SE}_m^2)$ is the harmonic mean of the variances of the individual point estimates.

Note the close connection between the standard errors for $\hat{\theta}_{\text{avg}}$ and $\hat{\theta}_{ivw}$. They both have the form $A^{1/2}/\sqrt{m}$, where A represents a mean of the SE_i^2. For simple averaging, the mean that appears in the standard error is the arithmetic mean, while for inverse variance weighted averaging, the harmonic mean appears. Since the harmonic mean is always less than or equal to the arithmetic mean, the inverse variance weighted average $\hat{\theta}_{\text{ivw}}$ is always the more efficient of the two.

6.1.5.2 Accounting for Uncertainty in the Standard Errors

In much of statistics, we often downplay the reality that standard errors are estimated rather than exact. For sufficiently large samples this may be fine, but in some settings where data integration is to be carried out, the individual estimates may be based on small samples, and/or the number of estimates to be integrated may be small. For this reason, it is appropriate to consider the consequences of using plug-in variance estimates in the procedures discussed above.

A common way to quantify estimation imprecision in a standard error is through the *degrees of freedom*, which we denote k here (or k_j for the degrees of freedom of the j^{th} subpopulation). As the degrees of freedom increases, the estimated standard error becomes more accurate. In many basic example, the degrees of freedom is the sample size minus the number of parameters estimated for the effect of interest itself. For example, suppose our goal is to estimate and conduct inference for the sample mean. The standard error of the mean (SEM) is $\sigma/\sqrt{n}$, and n is typically known. Since σ must be estimated, we will use the same data to estimate σ as we used to estimate the mean. Since the mean is one parameter, there are $n-1$ degrees of freedom remaining for estimating σ. This is just one example, but the idea is broadly applicable. It turns out that an expression of the form $\widetilde{\text{SE}}_{\text{ivw}} = \sqrt{H(1+F)}/\sqrt{m}$ can be used as an approximate standard error for $\hat{\theta}_{\text{ivw}}$ that reflects the additional uncertainty resulting from estimation of per-parameter variances and standard errors. Here, F is the inflation term

$$F = 4m^{-2}H^2 \sum_j w_j(m/H - w_j)/k'_j,$$

where $k'_j = k_j - 4(m-2)/(m-1)$. Since $m/H = \sum w_i$, F is non-negative. When using $\hat{\theta}_{\text{ivw}}$ and this "second order" standard error, the analysis can be conducted using degrees of freedom $1/\sum_j(w_j^2/k_j)$, i.e. $\hat{\theta}_{\text{ivw}}/\widetilde{\text{SE}}_{\text{ivw}}$ can be taken to approximately follow a Student's t-distribution with this degrees of freedom.

6.1.5.3 Pooling Estimates that Cannot be Averaged

For most estimators, averaging is a sensible way to combine evidence from multiple sources, but there are a few cases where this is not a meaningful way to proceed. One example of this would be when the statistics of interest are quantiles. Suppose we have m subpopulations and θ_i is the p^{th} quantile of the outcome in the i^{th} subpopulation. The average of the θ_i does not correspond to a meaningful pooled estimate, since it is not the p^{th} quantile of the pooled population.

In fact it turns out that in order to pool the quantiles, we need the subpopulation-specific quantiles $\theta_i(p')$ at every probability point p', not only at $p' = p$. To pool quantiles, we exploit the fact that the solution to the equation

$$\int m^{-1} \sum_i \mathcal{I}(\theta_i(r) \le q)dr = p,$$

as a function of q, yields the pooled p^{th} quantile. To gain some intuition about this fact, consider the special case where $m = 1$, so that we have $\int \mathcal{I}(\theta(r) \le q)dr = p$. Applying the change of variables $u = \theta(r)$ we obtain $\int_{-\infty}^{+infty} \mathcal{I}(u \le q)f(u)du = p$, where f is the density function (assuming that one exists). From this expression it is clear that the values of q, p satisfying this equation are the quantile q and the corresponding probability point p.

To implement this approach, we replace the population quantiles $\theta_i(r)$ with their estimates $\hat{\theta}_i(r)$. It would not be possible in practice to estimate these quantiles at all values of r, so the integral is approximated with a finite sum, e.g. using the trapezoidal method, to obtain an estimating equation that can be evaluated in practice.

6.1.6 Assessment Instrument

- In one or two sentences, define a research aim that could be addressed using subpopulation estimates of birth weights, where the subpopulations are defined in a way that you specify.
- Write a roughly half-page memo to yourself identifying one or two key challenges that would arise when attempting to resolve your research aim, and indicating how these challenges could be overcome using methods discussed in this unit.

6.2 Assessing Heterogeneity

Time: 1 hour

Heterogeneity in statistics is mainly used to refer to situations where the structural relationships among measured or latent quantities vary across the subpopulations being assessed. This is not variation in the observations themselves, which vary according to a probability distribution determined by measurement errors and intrinsic variation. Instead, it refers to a deeper level of variation that is governed by all of the underlying factors, often unknown, that characterize the different subpopulations.

In this section, we will discuss ways to define heterogeneity in different settings, and how to estimate it from data. The major challenge will be to distinguish the variation of the two types discussed above – the inter-individual variation driven by measurement error and individual differences, and the heterogeneity driven by structural subpopulation differences.

6.2.1 Learning Objectives

1. Identify opportunities to apply visualization methods from applied statistics to the settings of meta-analysis and data integration.
2. Articulate the difference between statistical uncertainty and effect heterogeneity.
3. Explain the basis and purpose for the intraclass correlation coefficient, and contrast it with τ^2, the variance of true parameter values across subpopulations.
4. Explain how the law of total variation partitions total variation into the explained and unexplained fractions of variability.
5. Identify at least one setting where pooling by weighted averaging is inappropriate, and explain a more appropriate way to pool evidence in this setting.

6.2.2 Visualization-Based Approaches to Exploring Heterogeneity

The main distinguishing feature of aggregated data is that some of the variation is due to structural variation among the subpopulations, and some is due to imprecision in the subpopulation-specific estimates. But it is also important to remember that aggregated data are just data, and we do not always have to use specialized methods for data integration that focus on distinguishing these multiple levels of variation. Many descriptive and graphical approaches can be productively applied for understanding aggregated data.

We discuss just one possibility here, aiming to incorporate as much information as possible into a two-dimensional graph, influenced by the strategies of optimal scaling and biplots. Suppose we have two summary statistics S_i and T_i computed from the observations in the i^{th} subpopulation. A basic example would be to let S_i be the sample mean and T_i be the sample standard deviation. Technically these values are estimates with a non-zero standard error, but we ignore this fact here. Each subpopulation can be identified with the point (S_i, T_i) in the two-dimensional plane.

The i^{th} subpopulation is characterized by a set of categorical factors; for each level L_{jk} of factor j, where k indexes the levels, we can consider all the subpopulations i that have this level. We then take the weighted centroid of their (S_i, T_i) values, and plot a label for variable j and level L_{jk} at this coordinate. This is a type of dimension reduction or variable scaling that allows us to see how the subpopulations with specific attributes (defined through the factors) are distinct in terms of the outcome variable (defined through the two summary statistics).

6.2.3 A Two-Level Model for Subpopulation Variation

Suppose that we have data y_{ij} grouped into subpopulations, where i indexes the subpopulations and j indexes the units within a subpopulation. The subpopulations may be *unbalanced*, so that $j = 1, \ldots, n_i$ where n_i is the number of observations sampled from the i^{th} subpopulation, with the n_i varying. As an example, in the NCHS birth weight data y_{ij} could represent the birth weight of the j^{th} baby born in the i^{th} county, and n_i is the number of babies born in the i^{th} county.

A generative model for this situation is the one-way random effects model $y_{ij} = \mu + a_i + \epsilon_{ij}$, where $a_1, \ldots, a_m$ are random effects following a distribution with mean 0 and variance τ^2, and the ϵ_{ij} follow a distribution with mean 0 and variance σ^2. The scalar parameter μ is the overall mean, and the $\{a_i\}$ and $\{\epsilon_{ij}\}$ are mutually independent.

6.2.3.1 Relative Measures of Heterogeneity

In the one-way random effects model, τ^2 represents the heterogeneity, i.e. the variation between subpopulations, while σ^2 represents measurement error and other forms of inter-individual variation that differs from person to person. The ratio $\tau^2/(\tau^2 + \sigma^2)$ is called the *intraclass correlation coefficient*, or ICC. It falls between

0 and 1, and represents the fraction of the overall variation that is between rather than within subpopulations. The ICC is *equivariant* (it is invariant to translating and rescaling the data) and hence is dimension-free.

A simple but natural way to estimate the ICC is to estimate τ^2 as the sample variance of the subpopulation means, i.e. $\widehat{\text{var}}(\{\bar{y}_{i\cdot}\})$. This will tend to overestimate τ^2, since it conflates random errors in the sample means with structural variation in the a_i. If the subpopulation sample sizes are large relative to the within-subpopulation variance, this overestimation is minor. A more sophisticated approach for estimating the ICC uses linear mixed effects regression. While this approach has some advantages, it also has drawbacks, and we will not discuss it further here.

The notion of the ICC can be generalized to any statistic θ_i that can be defined at the level of the subpopulations. Suppose that corresponding to the true θ_i, we also have unbiased estimates $\hat{\theta}_i$ and estimated standard errors $\widehat{\text{SE}}_i$. In the case of the ICC, $\theta_i = E[y_{ij}]$ (for any j), $\hat{\theta}_i = \bar{y}_{i\cdot}$, and $\widehat{\text{SE}}_i = \widehat{\text{SD}}(\{y_{ij}\})/\sqrt{n_i}$, but here we allow the statistic to be essentially anything, not exclusively the sample mean.

6.2.3.2 Absolute Measures of Heterogeneity

The *law of total variation* is a powerful tool for isolating the structural variation of a statistic from the variation attributable to other factors such as sampling and measurement variation. If the statistics are unbiased so that $E\hat{\theta}_i = \theta_i$, then we have

$$\begin{aligned} \text{Var}(\hat{\theta}_i) &= \text{var}E[\hat{\theta}_i|\theta_i] + E\text{var}(\hat{\theta}_i|\theta_i) \\ &= \text{var}(\theta_i) + E[\text{SE}(\hat{\theta}_i)^2]. \end{aligned}$$

This relationship shows us that the variance among the estimates $\hat{\theta}_i$ is the sum of the variance among the true statistics and the average estimation variance. As long as our statistics are unbiased and we can calculate their standard errors, this result can be directly applied.

Applying this result to the setting of subpopulation means, we have that the variance of the θ_i can be estimated as

$$\hat{\tau}^2 = \widehat{\text{var}}(\bar{y}_{i\cdot}) - \text{Avg}_i\widehat{\text{SE}}_i^2.$$

This is an unbiased method of moments estimate of $\tau^2 = \text{var}(\theta_i) \geq 0$. However the estimator $\hat{\tau}^2$ can be negative; in fact no unbiased estimator exists that is always non-negative. This is not a major problem in most cases, as a negative estimate indicates that τ^2 is likely to be very small (we will consider approaches for quantifying uncertainty in the estimate below). The value of $\hat{\tau} = \sqrt{\hat{\tau}^2 \vee 0}$ has the same units as the data y_{ij}, making it more interpretable, but it is not equivariant. Lack of equivariance is not always a problem, and this "corrected" estimate can be used to form a corrected ICC, which is equivariant, by plugging it into the expression $\tau^2/(\tau^2 + \sigma^2)$.

6.2.4 Assessment Instrument

- In one or two sentences, propose a brief research aim for the NCHS data centering on heterogeneity.
- Write a brief reflection discussing an analytic approach that you would employ to resolve your research aim, discussing at least one potential challenge that is likely to arise and how you would overcome it.

6.3 Modern and Robust Approaches to Evidence Combination

Time: 1 hour

This section discusses more modern and advanced methods for data integration. These approaches are more conceptually advanced than the methods discussed earlier, but may yield deeper or more robust results, especially in more challenging settings.

6.3.1 Learning Objectives

1. Articulate the purpose and main idea behind empirical likelihood approaches for data integration.
2. Explain and contrast the definitions of p-values and E-values, and identify some advantages and effective uses for each.
3. Explain the rationale for partitioning variance using estimated variance components in a meta-analysis setting, and explain the meanings of explained variation, random main effects, and idiosyncratic variation.

6.3.2 Empirical Likelihood Approaches to Heterogeneity

The method of moments approach to quantifying heterogeneity is simple to apply if the estimators are unbiased ($E\hat{\theta}_i = \theta_i$) and if we have a good way to calculate standard errors for the $\hat{\theta}_i$. If these conditions are difficult to meet, an alternative approach based on jackknifing can be employed. We illustrate the approach for estimation and inference of the between-subpopulation variance τ^2, but the idea would work for other estimands as well. Let $\hat{\tau}^2_{-i}$ denote the estimate of τ^2 excluding all observations in subpopulation i. The quantity $\eta_i \equiv m\hat{\tau}^2 - (m-1)\hat{\tau}^2_{-i}$, where m is the number of subpopulations, is known as the *pseudo-observation* for the i^{th} subpopulation. It approximately captures the unique contribution of the i^{th} subpopulation cluster to the estimate $\hat{\tau}^2$.

An *empirical likelihood* on the $\{\theta_i\}$ is a discrete probability distribution that places probability mass p_i on θ_i, where all $p_i \geq 0$ and $\sum_i p_i = 1$. Suppose we wish to test the null hypothesis that $\tau^2 = \theta$, for a given value θ. We can fit an empirical likelihood subject to the constraint $\sum_i p_i \eta_i = \theta$. We now have two equality constraints on the p_i, which is not sufficient to uniquely identify them, so we proceed by finding the distribution with maximum log-likelihood $\sum_i \log(p_i)$ that satisfies the two constraints. The objective function is concave and the constraints are linear, so this optimization problem has a unique solution $\hat{p}_i(\theta)$ for each θ.

We now proceed to consider how uncertainty assessments can be obtained in this setting. Let $L(\theta) = \sum_i \log(\hat{p}_i(\theta))$ denote the maximized likelihood value at θ, and let θ^* denote the argmax of this function. We can refer to $L(\theta)$ as the *profile likelihood*. In most cases, θ^* will be identical to or close to the full data estimator, which in our case is $\hat{\tau}^2$. To construct a confidence region for this estimate, we use test inversion. Since $D(\theta) \equiv -2(L(\theta) - L(\theta^*))$ follows a χ^2 distribution with 1 degree of freedom, we can include θ in the 95% confidence region iff $D(\theta) < Q$, where Q is the 0.95 quantile of the χ^2_1 distribution.

Another role for empirical likelihood in the setting of data integration is to allow for nonparametric estimation of the distribution of true subpopulation effects. As above, let θ_i denote the true mean (e.g. birth weight) for subpopulation i, with standard error $s_i = \sigma_i/\sqrt{n_i}$. We can take $\hat{\theta}_i$ to follow a normal distribution centered at θ_i, with standard deviation s_i. We then model the distribution of θ_i as being a finite distribution supported on a grid of r points $g_1, \ldots, g_r$, with unknown probability masses $p_j = P(g_j)$. The problem resembles a finite mixture where the unknown value of θ_i (one of the g_j) is an unobserved random variable which is marginally distributed on the g_j with probabilities p_j. EM or other algorithms are well-established for fitting such mixture models.

6.3.3 E-values

"Expectation values", referred to as E-values, are an emerging way to conduct statistical inference that aims to overcome some of the shortcomings of the dominant inferential approaches based on p-values. E-values have particular relevance in meta-analysis and data integration, since they allow meaningful pooling of evidence under weaker conditions than p-values. At the present time, E-values are still a rather new idea, and are not yet fully developed. E-values require a fairly distinct way of thinking compared to p-values, and there are not yet default or automatic ways to use E-values in all settings where p-values can be used. It is not realistic to view E-values as a replacement for p-values at the present time, but it is valuable to be aware of this new and promising framework. Here we will give an introduction and some elementary examples of inference using E-values, focusing on applications in data integration and meta-analysis. For more information, here are links to a comprehensive ebook and an overview paper.

Like other approaches to statistical inference based on p-values, Z-scores, and false discovery rates, E-values are used in the setting where we have an explicit null hypothesis, and a statistic used to measure evidence against the null hypothesis. The definition of an E-value is straightforward – it is a non-negative statistic whose expected value when the null hypothesis is true is less than or equal to 1. Recall that a p-value is a statistic that follows a uniform distribution on $[0, 1]$ (or stochastically dominates such a null distribution) when the null hypothesis is true, and a Z-score is a statistic that follows a standard normal distribution when the null hypothesis is true. Placing a condition on the expectation is much weaker than placing a condition on the distribution, therefore E-values by definition are less constrained than p-values and Z-scores. In many cases, this makes it easier to construct E-values, and the E-values are generally more robust to assumptions being violated because the requirements imposed on them are weaker.

When the null hypothesis is not true, it is desirable for E-values to tend to be larger than 1, and as more data are obtained, we can construct an E-process that meaningfully reflects the aggregation of evidence with increasing sample size. Thresholds on any evidence scale are somewhat arbitrary, but for E-values, one can take $1/E$ as an upper bound on the p-value, so it is reasonable to take $E \approx 20$ as minimal evidence against the null hypothesis, roughly comparable to $p < 0.05$, and $E \approx 100$ as strong evidence against the null, equivalent to or stronger than $p < 0.01$.

An easy way to get started with E-values is to use a "p-to-e calibrator", which converts p-values to E-values. The E-values constructed this way are usually not optimal so this tends not to be used much in practice, but it is a way to start building intuition. A basic calibrator is $(1-p+p\log(p))/(p\log(p)^2)$, which converts any valid p-value p to a valid E-value.

A common way to construct optimal E-values is using likelihood ratios. The simplest case is where we have a simple null hypothesis and a simple alternative hypothesis (recall that a simple hypothesis is one which contains a single probability distribution). In this case, the likelihood ratio is an E-value. If the null hypothesis is composite, we can simply plug in the MLE (under the null) of the nuisance parameters, just as in traditional likelihood ratio analysis. If the alternative hypothesis is composite, a somewhat more complex approach known as *reverse information projection* is used, but we will not discuss that further here.

Given a test statistic that is a Z-score, i.e. that follows a standard normal distribution when the null hypothesis is true, we can construct an E-value as $E = \exp(bZ)/c$. The parameters b and c are chosen so that the expected E value is no greater than 1 when the null hypothesis is true. Since Z is standard Gaussian, $\exp(bZ)$ is log-normal with mean equal to $\exp(b^2/2)$. Thus, $\exp(bZ - b^2/2)$ is an E-value. This approach can be used to produce E-values from many non-parametric procedures that yield asymptotically normal reference distributions when the null hypothesis is true.

Just as there are many ways of constructing a valid p-value for a given null hypothesis, there are many ways to construct valid E-values. In fact, since E-values are less restrictive than p-values, there are more ways to construct E-values than p-values for any given setting. To compare different E-values, we need a notion of power, analogous to the traditional notion of statistical power in hypothesis testing. In the context of E-values, people speak of "growth rates", to refer to the rate at which the E-process grows as the sample size increases. If one E-process always dominates another in terms of its growth, the latter is viewed as inadmissible. In the context of E-values constructed using the form $\exp(bZ)/c$, where Z is a Z-score, the constant c is used to impose the constraint that the expected E-value is 1. The constant b can be used to tune the power of the E-value. The value of b can be seen as determining the skewness of the E-value's null distribution, with larger values of b resulting in greater right skewness. We do not have a simple way to optimally set the value of b, but from experimentation it seems that neither perfect symmetry nor very strong right skew are optimal. Strong but not excessive right skew in the E-value's null distribution seems to perform best.

One of the important features of an E-value is that it allows "anytime valid inference". This is not directly applicable for the NCHS data, which as administrative data are available to us in entirety. But imagine a situation where we must collect data to assess the null hypothesis, and we wish to periodically check whether the data we have collected so far yield strong evidence against the null. In classical hypothesis testing, this introduces many challenges due to the bias introduced when stopping in a data-dependent manner. But with E-values it is perfectly valid to sequentially collect data until we reach a threshold, say $E \geq 20$, and stop collecting data at that point. We may still report our E-value as if we had planned ahead of time to stop at that particular point.

In meta-analysis, where researchers frequently re-assess the literature incorporating newer works along with the older evidence, anytime valid inference resolves potential concerns about overly-optimistic assessment of effects due to continuous reassessment of evidence. Furthermore, as indicated at the outset, E-values are only required to meet a fairly weak expectation constraint, which is much more robust to dependence compared to p-values. Network and cluster effects are well-documented in the scientific literature, so this is another reason that E-values may eventually become a powerful tool for meta-analysis and data integration, even if they are not widely used in practice today.

6.3.4 Stable and Idiosyncratic Variation

Variance component analysis is a special type of multilevel modeling that arises when all explanatory variables are categorical. We can use variance components analysis to gain insight into data that have been partitioned into subpopulations, especially when the subpopulations are defined through the intersection of several factors (e.g. year, sex, maternal race, location). This methodology allows us to rigorously and quantitatively assess the relationships between the factors defining subpopulations (e.g. year) and a quantitative response such as birth weight.

One way to approach variance components analysis is through fully-specified "generative" probability models, but the notation and language for describing more complex multilevel models gets quite complex. Most variance components analysis is conducted using restricted maximum likelihood (RMLE) methods that are supported in statistical packages such as the lme4 package in R, and in SAS, Stata, and Julia. Although the Python Statsmodels package can fit certain types of linear multilevel models, it lacks capabilities comparable to R. Moreover, there are reasons to consider alternative estimators to the common RMLE approach, since, among other issues, RMLE takes the random effects to be Gaussian, and it is unclear how violation of this modeling assumption impacts the results. In addition, it is difficult to account for complex patterns of heteroscedasticity using existing implementations such as lme4 in R. There are a number of alternative estimators for variance components models, including the MINQUE approach, as well as Bayesian approaches. Here we focus on one of these alternatives.

We will demonstrate how complex variance components parameters can be estimated using a method of moments approach based on U statistics. An advantage of this approach is that it allows us to directly target variance parameters of interest without explicitly specifying a complete model for the data. Also, it can accommodate arbitrary known heteroscedasticity. These estimates can be obtained quickly using just a few lines of code.

For illustration, suppose we have subpopulation means for subpopulations defined by location i, sex j, and maternal race k, with race coded as four levels. We can consider the following model

$$y_{ijk} = \mu + \beta_{2j} + \beta_{3k} + \theta_i + \theta_{ij} + \theta_{ik} + \epsilon_{ijk}, \tag{6.1}$$

where μ, the β_{2j}, and the β_{3k} are all *fixed effects*, while the θ_i, the θ_{ij}, and the θ_{ik} are all random effects, and the ϵ_{ijk} represent unexplained variation. Since the data are actually averages rather than individual observations, the ϵ_{ijk} are anticipated to be strongly heteroscedastic with (approximately) known variances, $\text{var}(\epsilon_{ijk}) = \sigma^2_{ijk}/n_{ijk}$, where σ^2_{ijk} is the variance within subpopulation ijk (which can be estimated in the usual way), and n_{ijk} is the sample size of subpopulation ijk. All random effects in model **??** are constrained to have expected value zero. As usual when fitting regression models with categorical explanatory variables, one level of each variable is treated as the reference level and its coefficient set to zero, e.g. $\beta_{21} = 0$ and $\beta_{31} = 0$. Importantly, this is done for the fixed effects but not for the random effects.

The model expressed in **??** is a specific example of an ANOVA-type decomposition. Its marginal mean structure is

$$E[y_{ijk}] = \mu + \beta_{2j} + \beta_{3k}. \tag{6.2}$$

This mean structure allows for stable differences between sexes, captured through the β_{2j}, and stable differences between races captured by the β_{3k}. By "stable difference" we mean a difference that is present on average across the random effects, holding the covariates responsible for fixed effects at specific values. For example, if $j = 2$ corresponds to female babies, then $\beta_{22} - \beta_{21}$ is the marginal mean difference in birth weights between female and male babies, holding race fixed (at any level), and averaging over the locations.

The mean structure **??** is not *saturated*; for example we do not include sex by race interactions. Specifically, it is *additive*, which means, for example, that the contrast $\beta_{22} - \beta_{21}$ is the marginal mean difference between female and male babies of the same race, regardless of what this race is. To allow the marginal mean sex differences to differ by race, we would need to include a sex by race interaction in the mean structure, but we do not pursue that direction here since we want to focus on the variance components structure. Since there are no interactions in our marginal mean structure, the β_{2j} and the β_{3k} can be referred to as *fixed main effects* for sex and maternal race, respectively.

We also note that we do not include fixed effects for location i. One reason for this is practical, as there are thousands of locations and the model would have a lot of parameters. On the other hand, a lot of modern statistical methodology considers high dimensional models with many parameters, so this is no longer a technical obstacle, and a modern computer could easily fit a model with fixed main effects for the thousands of locations. Another commonly expressed perspective is that we use random effects when the levels of a group are a random sample from a superpopulation, or when we don't "care about" specific levels of the variable. But these considerations are quite subjective. We take the perspective here that this is a modeling choice, and do not concern ourselves now with further justification for which variables are included as fixed effects.

The emphasis here is on the random effects, specifically the θ_i, the θ_{ij}, and the θ_{ik}. These are required to have mean zero, since any nonzero mean structure in the random effects would not be identifiable relative to the fixed effects. They are also required to be mutually independent. In terms of terminology, we can refer to the θ_i as random location "main effects", while the θ_{ij} are "idiosyncratic sex differences" within locations, and the θ_{ik} are "idiosyncratic race differences" within locations. The term "idiosyncratic" here emphasizes that, e.g. $\theta_{i2} - \theta_{i1}$ represents the female-to-male contrast uniquely within location i, after removing the marginal female-to-male contrast. To make the female-to-male mean difference $\beta_{22} - \beta_{11}$ identified and meaningful, we need the $\theta_{i2} - \theta_{i1}$ to have expected value zero, which follows from the fact that all random effects have zero population expectation.

Let y_{ijk} denote the sample mean of the residuals from an ordinary least squares fit of the mean structure model **??**. Let $\hat{\sigma}^2_{ijk}$ denote the sample variance of these residuals, and let n_{ijk} denote the number of observations in cell ijk. For any $j \neq j'$,

$$E[(y_{ijk} - y_{ij'k})^2] = 2\text{var}(\theta_{ij}) + \sigma^2_{ijk}/n_{ijk} + \sigma^2_{ij'k}/n_{ij'k}. \tag{6.3}$$

Thus

$$U_{ijj'k} \equiv ((y_{ijk} - y_{ij'k})^2 - \sigma^2_{ijk}/n_{ijk} - \sigma^2_{ij'k}/n_{ij'k})/2$$

is an unbiased estimate of $\text{var}(\theta_{ij})$. The final estimator of $\text{var}(\theta_{ij})$ is $\text{Avg}_{i,j,j',k}U_{ijj'k}$. One could use inverse variance weighting to improve the precision of the estimate, but we do not pursue that here.

6.3.5 Assessment Instrument

- Choose one of the three approaches discussed in this unit (jackknife empirical likelihood, E-values, or variance component analysis), and write a persuasive paragraph addressed to a clinical researcher, aiming to explain in a non-technical way the basis of the approach, and proposing a possible way for them to take advantage of this newer and more advanced methodology in their research.

6.4 Evaluation Rubric

Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
1. Research Aim & Key Challenges (20%)	Defines a clear, specific, and feasible research aim using subpopulation estimates of birth weights; the memo identifies two well-developed challenges directly tied to the aim; proposes concrete, method-grounded strategies from the unit to overcome each challenge; writing is precise and well-organized.	Research aim is clear and reasonable; memo identifies at least one well-developed challenge with a mostly adequate strategy; connection to unit methods is present with minor gaps in depth or specificity.	Research aim is present but vague or overly broad; memo identifies a challenge but treatment is superficial; strategies proposed are generic or only loosely connected to unit methods.	Research aim is missing, unclear, or not grounded in subpopulation estimates; memo is absent or does not meaningfully identify challenges; no connection to unit methods.
2. Heterogeneity: Research Aim & Reflection (20%)	Proposes a specific, well-motivated research aim centering on heterogeneity in the NCHS data; reflection clearly articulates the chosen analytical approach; identifies at least one substantive challenge with a thoughtful, method-specific strategy to overcome it; demonstrates nuanced understanding of the distinction between statistical uncertainty and effect heterogeneity.	Research aim is reasonable and centered on heterogeneity; reflection discusses the approach with adequate depth; at least one challenge identified with a mostly adequate strategy; understanding of uncertainty vs. heterogeneity mostly clear, with minor gaps.	Research aim mentions heterogeneity but lacks specificity; reflection is present but superficial; challenge acknowledged without substantive strategy; distinction between uncertainty and heterogeneity only partially demonstrated.	Research aim does not meaningfully engage with heterogeneity; reflection absent or minimal; no challenge identified or addressed; little evidence of understanding heterogeneity concepts.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
3. Persuasive Communication of Advanced Method (25%)	Clearly identifies one of the three advanced methods and explains its basis in genuinely non-technical language accessible to a clinical researcher; proposes a specific, realistic application of the method to the researcher's context; argument is persuasive, well-structured, and demonstrates deep understanding of the method's rationale and advantages.	Method is correctly identified and mostly explained in accessible language; proposed application is reasonable with minor gaps in specificity or persuasiveness; demonstrates solid understanding of the method with minor technical lapses.	Method is identified but explanation is partially technical or unclear for a non-specialist audience; application is generic or weakly connected to clinical research; understanding of the method is partial.	Method not clearly identified or explanation is incorrect; language is overly technical or inaccessible; no meaningful application proposed; little evidence of understanding the advanced methods.
4. Conceptual Mastery: p-values, E-values, & Error Control (20%)	Accurately explains and contrasts p-values and E-values, including their definitions, assumptions, and practical advantages; correctly distinguishes FPR, FWER, and FDR with concrete examples; clearly articulates the key ideas behind the local FDR approach; analysis is precise and demonstrates sophisticated conceptual understanding.	Explains and contrasts p-values and E-values with mostly accurate detail; distinguishes error control approaches with reasonable examples; local FDR explanation mostly correct with minor gaps.	Explains p-values and E-values at a basic level; distinguishes error control approaches without concrete examples or with imprecision; local FDR mentioned but not clearly explained.	Explanation of p-values or E-values is missing or incorrect; error control approaches conflated or absent; no meaningful engagement with local FDR.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
5. Overall Quality & Integration (15%)	All three responses are coherent, well-structured, and clearly connected to course concepts; writing is precise and appropriately tailored to the intended audience; demonstrates thorough command of data integration and meta-analysis concepts throughout.	Responses are mostly coherent and well-structured; course concepts applied correctly with minor lapses; writing is clear with occasional imprecision; good overall command of unit material.	Responses are partially coherent; some components feel disconnected or underdeveloped; writing is adequate; course concepts applied unevenly across the three tasks.	Responses are incoherent, incomplete, or largely disconnected from course content; writing is unclear; limited evidence of engagement with unit material.

Chapter 7

Unit 7: Large Language Models in Biomedical Research

Total Time: 5 hours

This unit prepares participants to use large language model (LLM) based AI agents as rigorous, accountable tools in biomedical data science research. The unit is *load-bearing*: its deliverables are components of the 2026 Data Challenge final report. Students work with the same NCHS vital statistics database used throughout the bootcamp, using an LLM ensemble to assist with data navigation, methodological decision-making, grant review, and manuscript preparation, while maintaining human accountability for every output.

Three principles organize the unit. First, invalidation (a framework broader than "hallucination" that encompasses factual, logical, normative, and structural breaches) is mathematically inevitable in any LLM-generated output; human verification is therefore not optional. Second, cross-agent critique reduces invalidation rates below what any single model achieves, but introduces its own ethical trade-offs that must be actively managed. Third, any sociodemographic variable whose encoding has changed across the historical record carries risks of structural and normative invalidation when queried without contextual grounding.

7.1 The Ethics of AI Agents

Time: 1 hour (Instructional: 50 minutes, Protocol Construction: 10 minutes)

This lesson examines the ethical dimensions of using LLM-based AI agents in biomedical data science. Unlike single-turn chatbot interactions, agentic workflows involve planning, tool use, persistent memory, and multi-agent coordination, substantially expanding the ethical surface area of AI-assisted research. Students develop a framework for identifying and classifying invalidation risks, analyze the specific ethical challenges posed by historical vital statistics data with inconsistently encoded racial and ethnic categories, and produce a *Research AI Governance Protocol* that governs their own use of LLM tools throughout Sections **??** and **??**.

The lesson proceeds in four instructional blocks:

1. **Agent architecture and expanded ethical surface (15 min).** What distinguishes an AI agent from a chatbot: planning, tool use, memory, and multi-agent coordination. Each capability adds a distinct ethical surface. The discursive network framework explains why agent-generated statements that enter manuscripts acquire authority independent of their accuracy, making verification a structural responsibility rather than an optional quality check.

2. **Invalidation: a taxonomy broader than hallucination (15 min).** The four types of invalidation (factual, logical, normative, and structural) are introduced using NCHS-specific examples. Factual: an LLM names a variable not present in the DVS schema as a named field. Logical: an LLM recommends a rate denominator inconsistent with the research question's population definition. Normative: an LLM interprets racial disparities in birthweight as reflecting biological differences between racial groups rather than the consequences of racism. Structural: an LLM queries APGAR score columns for records before 1977, when those columns did not exist in the flat file. Each type requires a distinct verification strategy.

3. **Sociodemographic consciousness in biomedical data science (15 min).** We explore how many sociodemographic variables are social construct with no valid biological basis, but their reification in law, medicine, and data systems makes it real in its consequences. Birth outcomes differ by recorded race not because of intrinsic racial biology but because of racism: differential access to prenatal care, chronic stress from discrimination, environmental exposures in historically redlined neighborhoods, and differential clinical treatment.

   The specific NCHS data students are working with encodes race through a system that changed across the study period. From 1969 through 1977, race was assigned from a fixed set (White, Black, American Indian, Chinese, Japanese, Other), and for parents of different recorded races the child's race followed the mother's. Assignment rules were revised in 1978, inconsistently across states. The "Hispanic" ethnic category appeared late in the period and was collapsed into "White" in some years. The `child_race_recode_3` field changed its encoding midway through the study period. Any LLM query about race in this dataset risks producing normative invalidation by treating these categories as stable, comparable, and biologically grounded across the full 1969–1986 span.

4. **Ethics checklist and governance protocol construction (20 min).** The seven risk categories from the ethics framework (epistemic, accountability and authorship, provenance and reproducibility, confidentiality, bias and fairness, security, and sustainability) are reviewed with emphasis on the tradeoffs specific to this bootcamp context. Multi-agent critique tradeoffs:

privacy amplification, epistemic homogenization, compute burden, and attribution diffusion. Students construct the Research AI Governance Protocol that governs their subsequent work.

7.1.1 Learning Objectives

1. Define an AI agent in terms of planning, tool use, memory, and multi-agent interaction, and explain how each capability expands the ethical surface area relative to single-turn LLM use.

2. Apply the four-type invalidation taxonomy (factual, logical, normative, and structural) to concrete scenarios involving the NCHS vital statistics database, and identify the appropriate verification strategy for each type.

3. Articulate why sociodemographic variables historical US vital statistics data often reflect social constructs whose reification has real health consequences, and explain the specific coding inconsistencies in the NCHS 1969–1986 dataset that make naive racial comparisons normatively invalid.

4. Construct a Research AI Governance Protocol specifying provenance logging, invalidation classification, sociodemographic variable handling, escalation thresholds, and a compliant disclosure statement for the team's subsequent AI-assisted work.

7.1.2 Assessment Instrument

Item 1 — Invalidation Taxonomy Application (in-session, 8 minutes): The instructor presents four NCHS-specific LLM output scenarios, one per invalidation type. For each scenario, teams identify: (a) the invalidation type, (b) why it constitutes that type and not another, and (c) the verification method that would detect and correct it. Scenarios are drawn from realistic ensemble outputs on the vital statistics database.

Item 2 — Research AI Governance Protocol (in-session construction, 10 minutes; referenced throughout Sections ?? and ??): Each team produces a one-page governance protocol containing all five required components:

1. **Provenance logging convention:** how prompts, model identifiers, and timestamps will be recorded for every ensemble interaction.

2. **Invalidation log format:** the template for recording invalidation type, description, and resolution for each detected LLM failure.

3. **Sociodemographic variable handling clause:** specific commitments about how sociodemographic variables with historically unstable encoding will be treated in LLM queries. LLM queries involving sociodemographic variables must inject the relevant coding history as context; temporal comparisons of category rates require explicit justification; normative claims about disparities must be flagged for human review.

4. **Escalation threshold:** the conditions under which LLM output will not be used without independent verification or correction (e.g., any factual claim about NCHS variable definitions; any normative statement about population health disparities; any generated code before execution).

5. **Draft disclosure statement:** a single sentence in the form specified by the ethics framework, naming tools and tasks, to be finalized at the end of Section **??**.

The governance protocol is a live document. It is updated throughout Sections **??** and **??** as invalidations are encountered and resolved, and its disclosure statement is finalized as the unit's final deliverable.

7.2 AI Agents for Technical Tasks

Time: 2 hours (Instructional: 40 minutes, Activity Work: 80 minutes)

This lesson teaches students to use an LLM ensemble as a rigorous technical collaborator on analytical tasks grounded in the NCHS vital statistics database. The session has four sequential blocks. Block A derives the mathematical basis for multi-agent consensus and produces the agent-count commitments that govern subsequent activities. Blocks B, C, and D form an analytical chain: outputs from each block feed the next, and their collective products are load-bearing components of the final data challenge report.

Students enter this session with a completed model report from Unit 4, which documents a predictive model for the data challenge outcome (underweight birth rate or infant mortality rate by Texas county) including the feature selection memo, model comparison table, residual diagnostics, and limitations section. This report is the primary document the LLM ensemble will be asked to interrogate. The governance protocol from Section **??** is active from the start of this session; every ensemble interaction concerning the Unit 4 model is logged in the invalidation log.

The governing framework is the *Flaws-of-Others* (FOO) algorithm, in which multiple agents independently produce outputs, each critiques the others' responses, and a harmonizer synthesizes the critiques into a refined result. The minimum number of mutually detecting agents required to reduce the long-run false-statement share below a tolerance ε is:

$$n_{\min} = \left\lceil 1 + \frac{p\left(\dfrac{1}{\varepsilon} - 1\right) - q}{d} \right\rceil \tag{7.1}$$

where p is the per-agent invalidation rate, q is the internal self-repair rate ($q = 0.05$ reference value), and d is the cross-detection probability ($d = 0.19$ reference value). Students use this formula to make defensible, quantitative decisions about ensemble size for each activity.

Block A — FOO Framework: Derivation and Calibration (20 minutes instructional)

The derivation proceeds from first principles. A single agent has an effective invalidation rate p (combining intrinsic corruption and fabrication). Internal self-repair operates at rate q. Each additional cross-checking agent adds detection hazard d to the effective correction rate. The steady-state false-statement share with n mutually detecting agents is:

$$\pi_F^{(n)} = \frac{p}{p + q + (n-1)d} \tag{7.2}$$

Setting $\pi_F^{(n)} \leq \varepsilon$ and solving for n yields Eq. (**??**). The invalidation floor (proved formally in the underlying theory) establishes that some error rate is mathematically inevitable in any finite-loss LLM; the FOO architecture is the engineering response to this impossibility result.

Students compute $n_{\min}$ for three task categories they will encounter in this session using the reference values above:

Task category	Example	p	ε	$n_{\min}$
Factual schema lookup	Identify correct byte range for birthweight	0.10	0.10	*compute*
Methodological judgment	Choose projection model for county-level rates	0.30	0.05	*compute*
Code generation	SQL query for core research question	0.20	0.05	*compute*

The computed values are entered into the governance protocol as the team's agent-count commitments for Blocks B, C, and D.

Block B — The Texas Births Problem (35 minutes)

Before the bootcamp, participants were asked to count live births in Texas in 1969 to mothers residing in Texas. Teams compare their answers. Historically, different correct implementations of this query return different counts. The task is to diagnose the source of disagreement using the LLM ensemble.

Teams query the ensemble with the NCHS data dictionary and ask it to: (a) identify every field relevant to the filter, (b) flag any ambiguities in the dictionary language, and (c) write a query implementing the most defensible interpretation. Teams execute the generated query and compare counts across the class.

The deliverable is annotated, executable code (Python or SQL) that implements the correct filter with comments explaining each disambiguation decision. This code is a candidate component of the reproducible pipeline developed in Unit 5.

Model report critique (15 min). Each team submits their Unit 4 model report to the ensemble and prompts it to identify any claim in the methods section that is (a) not reproducible from the written description alone, (b) in tension with a stated limitation, or (c) likely to require verification against the NCHS data dictionary. Teams record each flagged claim in the invalidation log with an invalidation type classification from Section ??. For each factual or structural invalidation identified, teams verify the ensemble's critique against the Unit 4 model code before accepting or rejecting it. Teams that find no invalidations in their Unit 4 report should examine whether the ensemble is producing false-negative outputs (a structural invalidation of the ensemble itself) and document their reasoning.

Block C — Sociodemographic Variable Integrity: Instructor-Led Exploration (15 minutes)

Building directly on Section ??, the instructor demonstrates how LLM behavior differs across three sociodemographic variables whose encoding changed across the NCHS study period. For each variable the instructor submits a common query to the ensemble, first without contextual grounding and then with the relevant coding history injected, and the class observes the difference in output.

The three demonstrations pursue the same comparative question: does the ensemble recognize that the variable is historically unstable, and does it flag its own normative or structural claims without prompting? Race is the variable most likely to surface normative invalidation; maternal education is most likely to surface structural invalidation arising from changes in coding categories; geographic identifiers are most likely to surface factual invalidation arising from county boundary or code revisions. Together they illustrate that the retrieval-augmented mitigation strategy from Section ?? generalizes to any variable with a documented encoding history.

Students take notes and update the sociodemographic variable handling clause in their governance protocol based on what they observe. The take-home deliverable (Assessment Item 5) asks them to apply the same comparative analysis independently to their own data challenge variables.

Block D — Methodological Decision Support (60 minutes)

Teams use the ensemble to interrogate the model family choice documented in their Unit 4 report. The core prompt pattern is: present the ensemble with the Unit 4 feature set, the outcome distribution description, and the model comparison table, then ask it to (a) identify which alternative model from the comparison was most appropriate given the stated distributional properties of the outcome, and (b) identify any model family that was not evaluated in the Unit 4 comparison but would be worth considering given the data structure. Teams apply Eq. (??) to determine the appropriate ensemble size for this task. Every ensemble recommendation is logged and verified against the Unit 4 model diagnostics before being accepted. If the ensemble recommends a model family that the student already evaluated and rejected in Unit 4, teams document whether the ensemble's reasoning accounts for the diagnostic evidence that motivated the rejection.

7.2.1 Learning Objectives

1. Derive the FOO minimum-agent formula from the steady-state false-share equation and compute n_{min} for tasks of varying epistemic difficulty, applying the result to make quantitative ensemble-size decisions.

2. Use an LLM ensemble to navigate an ambiguous biomedical data dictionary, verify outputs against the authoritative database schema, and produce annotated executable code that implements a defensible query with all disambiguations documented.

3. Recognize temporal instability and normative invalidation risk across sociodemographic variables in the NCHS dataset, using sociodemographic variables and apply retrieval-augmented mitigation strategies consistently across all three.

4. Produce a structured methodological consensus memo evaluating competing projection approaches, with FOO-calibrated confidence levels, a synthesis recommendation, and a retrospective comparison to the analytic plan developed in Unit 3.

7.2.2 Assessment Instrument

Item 3 — Verified Data Query: Texas Birth Count (Block B deliverable): Annotated, executable code implementing the correct Texas birth count filter. The annotation must explain: (a) each filter field used and why, (b) each source of ambiguity in the data dictionary and how it was resolved, (c) the count produced, and (d) at least one invalidation from the ensemble interaction, logged in the governance protocol format.

Item 4 — Invalidation Log (cumulative across Blocks B, C, and D, and updated through Section ??): A structured log of all LLM invalidations encountered during Unit 7. Each entry must record: invalidation type (factual, logical, normative, or structural), a one-sentence description of the failure, the resolution applied, and the section in which it occurred. The log must contain at least six entries across the full unit (minimum two from Section **??**). If a particular invalidation type does not appear in a team's experience, this must be explicitly documented with a note on what would have caused it to appear.

Item 5 — Sociodemographic Variable Integrity Memo (take-home, due before Session 7.3): A written memo of three to four paragraphs treating sociodemographic variables comparatively. For each variable the memo must specify: (a) which coding or classification changes affect the variable across the study period and their source in the data dictionary; (b) which temporal comparisons are analytically valid and which are not; (c) how the team's LLM queries will be structured to mitigate normative or structural invalidation specific to that variable; and (d) the updated sociodemographic variable handling clause for the governance protocol, incorporating all variables.

Item 6 — Methodological Consensus Memo (Block D deliverable): A structured memo containing: (a) a comparison table with one row per projection approach, recording ensemble consensus level (High/Medium/Low), key assumptions, major risks, and divergence points; (b) the team's synthesis recommendation with justification; (c) the FOO analysis, which approach generated the most divergent ensemble responses and what epistemic difficulty that signals; and (d) a retrospective comparison to the analytic plan designed in Section ??, noting whether the consensus confirms, complicates, or challenges the earlier design choices.

7.3 AI Agents for Grant Review and Manuscript Preparation

Time: 2 hours (Instructional: 20 minutes, Activity Work: 100 minutes)

This session transfers the technical skills developed in Section ?? to the most career-relevant applications of LLM-assisted research: grant review and manuscript preparation. Students work with canonical NIH grant proposals that include their actual panel summary statements, providing ground-truth verification for ensemble outputs. Students also prepare a Specific Aims page for a research proposal using Jackson Heart Study (JHS) data, applying the framework from Section ?? to the framing of health disparity research.

The governance protocol from Section ?? remains active. The special confidentiality concern of Section ??A is emphasized: federal agencies prohibit uploading unpublished proposal text to unapproved AI systems for grant review purposes. The proposals used here are canonical public examples; students are explicitly warned that this constraint applies to any future peer review panel work. The invalidation log from Section ?? is updated throughout.

Block A — NIH Grant Review with Ground-Truth Comparison (60 minutes)

Each team selects one canonical NIH grant proposal from the provided collection. These proposals are publicly available examples accompanied by their actual summary statements from review panels.

Prompt design (10 min). Teams design separate prompts for each of the five NIH review criteria: Significance, Investigator(s), Innovation, Approach, and Environment. Each criterion requires different domain reasoning; a single prompt for all five produces superficial output. Teams apply Eq. (??) per criterion, recognizing that Approach typically warrants a higher agent count than Environment.

Ensemble review (25 min). Teams run the consensus pipeline across all five criteria. The governance protocol is active; every ensemble interaction is logged.

Comparison analysis (20 min). Teams compare the ensemble's consensus review to the actual panel summary statement across all five criteria, recording: consensus score, panel score, strengths identified by each, divergence type (missing context, conservative bias, detected concern missed by panel, domain-norm gap), and agreement level.

Critical synthesis (5 min). Teams identify three empirical categories from their comparison data: what the ensemble reliably detected, what it missed systematically, and what varied case by case. This analysis informs a critical reflection on appropriate LLM use in grant preparation.

Block B — JHS Specific Aims: Ensemble-Assisted Drafting (60 minutes)

The coordinator for JHS activities provides: a specific JHS research question, the relevant JHS variables, and two to three published JHS papers as grounding context.

Prompt construction (10 min). Teams design a structured prompt that injects JHS context as a system prompt. The task framing is explicit: not "write my Specific Aims" but rather "given this JHS research question, this prior work, and these three candidate aims, evaluate which aim structure is most scientifically defensible and propose the structure with highest consensus confidence, flagging any claims about JHS data or methods that require independent verification against the provided literature."

Ensemble drafting (25 min). Teams run the FOO pipeline with JHS context. Every LLM claim about JHS data, published results, or methods is flagged for verification against the provided papers and logged in the invalidation log.

Human revision and sociodemographic framing review (20 min). Teams revise the LLM draft, correcting all logged invalidations. A mandatory step: teams explicitly evaluate whether the drafted Specific Aims treats sociodemographic factors as social determinants of health operating through identifiable structural pathways rather than as fixed biological or demographic properties of the study population. The review must confirm that the framing aligns with current NIH policy on sociodemographic variables in grant applications. Each revision is documented in the invalidation log.

Disclosure statement finalization (5 min). The draft disclosure statement from Section **??** is revised to reflect the actual scope and nature of AI assistance across all three sections. Teams select the appropriate disclosure level (minimal, standard, or detailed) and produce a finalized statement suitable for inclusion in a peer-reviewed publication.

7.3.1 Learning Objectives

1. Design criterion-specific prompts for LLM-assisted NIH grant evaluation; compare ensemble outputs to actual expert panel judgments; and characterize empirical patterns of agreement, divergence, and systematic miss.

2. Construct an ensemble-assisted Specific Aims draft for a JHS research proposal, verifying all LLM claims against published JHS literature and ensuring that sociodemographic factors are framed as social determinants of health

operating through structural pathways rather than as fixed biological or demographic properties.

3. Produce a finalized disclosure statement appropriate for inclusion in a peer-reviewed publication, accurately reflecting the scope of AI assistance and maintaining full human accountability for all outputs.

7.3.2 Assessment Instrument

Item 7 — NIH Grant Review Analysis (Block A deliverable): A structured comparison document containing: (a) the comparison table across all five NIH criteria with scores, identified strengths, divergence types, and agreement levels; (b) three empirical categories (reliable detections, systematic misses, and variable outcomes) derived from the comparison data with at least one concrete example per category; and (c) a one-paragraph critical reflection on appropriate LLM use in grant preparation, grounded in the evidence from the comparison rather than prior assumptions.

Item 8 — JHS Specific Aims Draft (Block B deliverable): A draft Specific Aims page produced through the ensemble pipeline and revised by the team, with: (a) all LLM invalidations documented in the invalidation log with verification sources cited; (b) a written framing review note confirming that sociodemographic factors are framed as social determinants of health operating through structural pathways are treated consistently with the framework from Section **??**, and that the overall framing is consistent with NIH policy; and (c) a brief annotation for each aim identifying which elements are human-authored and which were substantially shaped by ensemble output, consistent with the governance protocol.

Item 9 — Final Disclosure Statement (Block B deliverable, unit-closing): A single, finalized disclosure statement appropriate for inclusion in a methods section or acknowledgments section of a peer-reviewed publication. The statement must: name the tools and approximate versions used; specify the tasks for which AI assistance was employed across the full unit; describe the verification procedures applied; and include an explicit accountability statement affirming that all claims, analyses, and conclusions were reviewed and affirmed by the team, who take full responsibility for the work. The statement must also specify whether and how the LLM ensemble was used to review or revise the predictive model report produced in Unit 4, including a description of the verification procedure applied to ensemble critiques of that report. The statement must be consistent with the completed invalidation log.

7.4 Evaluation Rubric

Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
1. Invalidation Taxonomy Application (10%)	Correctly identifies all four invalidation types from the presented NCHS scenarios; provides precise, specific justification for each classification that demonstrates deep understanding of why the type applies and not another; proposes a concrete, type-appropriate verification method for each scenario; connects normative invalidation to the sociodemographic consciousness framework.	Correctly identifies most invalidation types; justifications are mostly accurate with minor gaps in specificity; verification methods proposed are appropriate with minor omissions; some connection to normative/sociodemographic consciousness dimension.	Identifies some invalidation types correctly but confuses at least one; justifications present but imprecise or incomplete; verification methods generic rather than type-specific; sociodemographic consciousness connection absent or superficial.	Fails to correctly identify multiple invalidation types or conflates types; justifications missing or incorrect; no meaningful verification strategies proposed; no engagement with normative dimension.
2. Research AI Governance Protocol (15%)	All five required components are present, complete, and specific to the NCHS/vital statistics context; provenance logging convention is operationalizable as written; sociodemographic variable handling clause addresses coding history, temporal comparisons, and normative claim flagging for sociodemographic variables as primary case and at least one additional variable; escalation thresholds are concrete and verifiable; disclosure statement follows the prescribed format and is appropriately specific. Protocol was demonstrably active and updated through Sections 7.2 and 7.3.	All five components present; most are specific and operationalizable with minor gaps; sociodemographic variable handling clause addresses most key concerns; some evidence of protocol use in subsequent sections.	Most components present but one or more are generic or incomplete; sociodemographic variable handling clause present but does not address temporal inconsistency or normative flagging substantively; limited evidence of protocol use beyond Section 7.1.	One or more required components missing; present components are too generic to operationalize; sociodemographic variable handling clause absent or addresses only surface concerns; protocol not demonstrably used in subsequent activities.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
3. Verified Data Query: Texas Birth Count (10%)	Code is executable and produces the correct count; all ambiguities in the data dictionary are identified and resolved with explicit justification; annotations clearly explain each filter field and each disambiguation decision; at least one ensemble invalidation is documented in governance-protocol format; code structure is compatible with Unit 5 reproducible pipeline conventions.	Code is executable and produces a defensible count with minor filter issues; most ambiguities identified and resolved; annotations present for major decisions; at least one invalidation documented.	Code is present but contains a filter error or produces an uncorrected count; some ambiguities identified but not all resolved; annotations incomplete; invalidation documentation minimal.	Code is absent, non-executable, or implements an incorrect filter without acknowledgment; ambiguities in the data dictionary not identified; no invalidation documented.
4. Unit 4 Model Report Critique (10%)	Ensemble critique is elicited with a well-designed prompt that directs attention to reproducibility, internal consistency, and data dictionary alignment; all flagged claims are classified by invalidation type with specific justification; each factual or structural invalidation is verified against Unit 4 code before acceptance; at least one false-negative or false-positive ensemble output is identified and documented in governance-protocol format; the invalidation log is updated consistently.	Critique prompt is adequate; most flagged claims classified by type with reasonable justification; verification performed for most invalidations; invalidation log updated.	Critique prompt is generic or insufficiently specific to the Unit 4 report; classification of flagged claims by type is partial or inconsistent; verification performed for some invalidations; log updated incompletely.	No critique prompt submitted, or critique not connected to the Unit 4 report; no classification by invalidation type; no verification against Unit 4 code; invalidation log not updated.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
5. Invalidation Log (10%)	Log contains at least six entries across the full unit with at least two from Section 7.2; all four invalidation types are represented or absent types are explicitly accounted for with a note on what would cause each to appear; every entry includes type, description, resolution, and section; log is consistent with the governance protocol format and with the disclosure statement.	Log contains at least six entries; most types represented or accounted for; most entries complete; minor inconsistency with governance protocol format.	Log contains fewer than six entries or multiple entries are incomplete; at least two invalidation types present; some entries lack type, description, or resolution.	Log contains fewer than three entries; entries are incomplete or mis-classified; absent types not accounted for; log is not consistent with governance protocol.
6. Sociodemographic Variable Integrity Memo (10%)	Memo addresses all sociodemographic variables with data-dictionary-grounded analysis; for each variable, temporal validity is specified with reference to the documented coding changes; LLM query mitigation strategy is concretely articulated per variable; governance protocol clause is updated and consistent across all three; sociodemographic variables are addressed with comparable analytical specificity.	All three variables addressed with mostly adequate grounding; temporal validity discussed with minor gaps for one variable; mitigation strategies present but not fully operationalized for all three; protocol clause updated; sociodemographic variables treatment mostly sound.	All three variables named but treatment is uneven; temporal validity addressed for sociodemographic variables but not fully for the other two; mitigation strategies generic rather than variable-specific; protocol clause minimally updated.	One or more variables absent or addressed without data-dictionary grounding; temporal coding inconsistency not addressed for at least one variable; no variable-specific mitigation strategy; sociodemographic variables treated as stable, ahistorical categories without acknowledgment of encoding changes.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
7. Methodological Consensus Memo (10%)	Comparison table is complete for all three approaches with accurate, specific entries for consensus level, assumptions, risks, and divergence points; synthesis recommendation is well-reasoned and explicitly connected to the data challenge context; FOO analysis identifies the most epistemically difficult approach and explains why divergence signals this; retrospective comparison to the Unit 3 analytic plan is substantive and either confirms or productively complicates the earlier design.	Table complete with mostly accurate entries; synthesis recommendation reasonable with adequate justification; FOO analysis present but lacks depth; retrospective comparison to Unit 3 present with minor gaps.	Table incomplete or contains inaccuracies for one approach; synthesis recommendation present but weakly justified; FOO analysis minimal; retrospective comparison to Unit 3 superficial or absent.	Table missing one or more approaches or largely inaccurate; no synthesis recommendation or recommendation is unjustified; FOO analysis absent; no retrospective connection to Unit 3.
8. NIH Grant Review Analysis (10%)	Comparison table covers all five criteria with accurate, evidence-grounded entries; three empirical categories (reliable detections, systematic misses, variable outcomes) are clearly distinguished with at least one concrete example each; critical reflection is grounded in comparison evidence rather than prior assumptions and reaches a nuanced, defensible conclusion about appropriate LLM use in grant preparation.	Table covers all five criteria with mostly accurate entries; three categories identified with examples, some less concrete; reflection grounded in evidence with minor gaps in nuance.	Table covers most criteria; categories identified but blurred or with weak examples; reflection present but relies on assertion more than evidence.	Table incomplete or inaccurate for multiple criteria; categories not meaningfully distinguished; reflection absent or disconnected from comparison evidence.

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Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
9. JHS Specific Aims Draft (10%)	Draft is complete (opening paragraph, three aims, closing paragraph); all LLM invalidations are logged with verification sources; sociodemographic review note confirms appropriate social-determinant framing and NIH policy alignment with specific evidence from the draft; annotation accurately distinguishes human-authored from ensemble-shaped elements; draft is scientifically coherent and grounded in the provided JHS literature.	Draft is mostly complete; most invalidations logged with sources; sociodemographic framing mostly appropriate with minor lapses; annotation present with minor gaps; draft is scientifically reasonable.	Draft has structural gaps (missing aim or section); invalidation log incomplete or sources missing; sociodemographic framing partially addressed without explicit review note; annotation minimal.	Draft is incomplete or incoherent; invalidation log absent or has fewer than two entries; no sociodemographic review performed; no annotation distinguishing human from ensemble contributions.
10. Final Disclosure Statement (5%)	Statement names tools and approximate versions; accurately specifies the tasks for which AI assistance was employed across all three sections; describes the verification procedures applied in terms consistent with the invalidation log; includes an explicit accountability statement affirming full human responsibility; is appropriate for inclusion in a peer-reviewed methods or acknowledgments section without revision.	Statement names tools and tasks with minor omissions; verification procedures described at a general level; accountability statement present; suitable for publication with minor editing.	Statement present but omits tools or tasks; verification described only vaguely; accountability statement present but formulaic; would require substantial editing for publication.	Statement absent, or present but does not name tools, tasks, or verification; no accountability statement; not suitable for publication in any form.

Chapter 8

Unit 8: Jackson Heart Study

Time: 3 hours

8.1 Introduction to the Jackson Heart Study

This lesson provides an overview of the Jackson Heart Study (JHS), focusing on its design, data collection, and variable interpretation. Students will describe the JHS's purpose, population, and exam structure, summarize clinical, survey, and genetic data collection methods across study phases, and learn to use JHS codebooks to identify variables for research questions.

8.1.1 Learning Objectives

1. Describe the JHS Study Design: Explain the purpose, population, and structure of the JHS, including its major exams.
2. Summarize Data Collection Methods: Identify the types of data collected (e.g., clinical, survey, genetic) and the methods used in different study phases.
3. Interpret Key Variables and Codebooks: Understand how to use JHS codebooks to find variables relevant to specific research questions.

8.1.2 Assessment Instrument

1. Describe in your own words the purpose of the JHS, cohort characteristics, and exam waves.
2. Describe how JHS collects specific data (e.g., CAC scores, lipid tests, etc.) and potential biases in data collection.
3. Answer the following question: "Association between hysterectomy and cardiovascular disease in the JHS, adjusting for covariates" by locating relevant variables in the JHS codebook. Describe the variables chosen and their rationale.

8.2 The Process of Manuscript Development in the JHS

This lesson covers the process for requesting and obtaining JHS data. Students will learn to navigate data access procedures, including submitting manuscript or ancillary study proposals, completing data use agreements, and addressing ethical considerations to ensure responsible use of JHS data.

8.2.1 Learning Objectives

Explain Data Access Procedures: Describe the process for requesting and obtaining JHS data, including data use agreements and ethical considerations.

8.2.2 Assessment

1. In your own words, enumerate the steps involved in the process for developing a JHS manuscript.
2. Using the information acquired from the lecture, draft a mock JHS manuscript proposal, using the Manuscript Proposal Form provided and the sample manuscript proposal.

8.3 Evaluation Rubric

Component	Excellent (90–100%)	Good (80–89%)	Satisfactory (70–79%)	Needs Improvement (<70%)
1. Variable Identification (50%)	Accurately identifies variables; clear rationale; demonstrates deep understanding of JHS datasets.	Minor omissions but overall strong alignment.	Adequate but limited justification.	Incomplete or incorrect variable selection.
2. Manuscript Proposal (50%)	Logically extends grant proposal; well-articulated and publication-ready.	Solid but needs refinement in framing or methods.	Basic understanding but lacks clarity or completeness.	Unclear, inconsistent, or incomplete proposal.
3. Overall Professionalism and Clarity	Polished writing, correct formatting, adherence to instructions.	Minor issues with organization or formatting.	Understandable but contains stylistic or format errors.	Poor structure, numerous errors.